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Knowledge graphs and large language models: Construction paradigms, integration approaches, and emerging applications

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ABSTRACT

Large Language Models (LLMs) have shown impressive abilities in natural language understanding and generation that have been widely used in various applications like content creation, answer generation, Q&A systems and have shown impressive capabilities in processing knowledge. The attention mechanism aids transformer-based models, like ChatGPT, to articulate well-structured and grammatically sound sentences. Nevertheless, these models are prone to hallucinations lacking fact-based information and structured reasoning. In contrast, Knowledge Graphs (KGs) are semantically rich graph-based data structures which are interconnected facts represented as entities (nodes) and relations (edges) enhancing factual grounding and explainability in LLMs. These semantic networks store structured and organized information enabling explicit modeling of domain knowledge. This survey presents a systematic review that aims at synthesizing KGs and LLMs. We present a detailed overview on Knowledge Graph construction, Multi-modal KG construction, Domain-specific KG construction with a focus on healthcare. Additionally, we classify integration strategies in three ways, namely, 1) KG enhanced LLMs, 2) LLMs-augmented KGs and 3) Synergized LLMs+KGs, the neuro symbolic architectures. We discuss recent research developments like HybridRAG and GraphRAG based on graph-based retrieval-augmented generation. We emphasized evaluation metrics and real-world applications in enterprise and clinical domains. We conclude by discussing open challenges in cross-modal alignment, dynamic knowledge updates and multi-hop reasoning. We outline future research directions towards reliable knowledge-grounded intelligent systems. We foresee that this research provides deeper insights for future research initiatives in the field of knowledge graphs.

1. Introduction

Large Language Models (LLMs) have shown remarkable advancements in artificial intelligence and are the default interface for information access, summarizations and code explanations across various domains. LLMs, such as OpenAI's ChatGPT, GPT-3 [1], Google's Gemini [2], and Anthropic's Claude [3] with billions of parameters are exceptional at text summarization and interpreting Human-like responses. Also, when integrated with Retrieval Augmented Generation (RAG) [4], LLMs use external vector databases to reduce hallucinations to keep themselves updated with real-time data. These capabilities of LLMs have shown remarkable innovations across industries such as e-commerce, healthcare, education, finance, telecommunications to public services. However, their generative strength has been criticized for inconsistent multi-hop reasoning, hallucinated facts, a lack of domain knowledge, and generating statements that are plausibly incorrect. LLMs generate answers using the knowledge contained in the training corpus. For example, LLMs, when asked, "Who discovered Radium?", could say, "Marie Curie discovered Uranium in 1898", which violates the fact that Marie Curie discovered Radium. This adversely affects the reliability of LLMs. The presence of hallucinations in pre-trained language models makes it difficult for users to distinguish between accurate reasoning and fabricated content. These limitations highlight the need for advanced fact-verification layers, such as knowledge graph alignment, to uphold model reliability. Retrieval Augmented Generation (RAG) [4] acts as an alternative to incorporate knowledge from external databases. Particularly known for knowledge-intensive tasks, it proposes a fine-tuning process which combines parametric and non-parametric memory mechanisms to generate coherent outputs. Yet, RAG performs retrieval based on local relevance; it lacks awareness of the entire dataset's global patterns. To overcome this problem, there has been an evolution in recent times. We witnessed a rise of graph-based variant,

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GraphRAG [5,137], which has gained significant attention for its ability to enhance the Q&A generation and extend RAG by using graph-based community detection for enhanced multi-hop retrieval (discussed in detail in Section 8.1.2).

Knowledge Graphs (KGs) excel in harnessing the large and interconnected data that has been diversified across the globe. Knowledge representation tools, when integrated with LLMs, boost their performance, making LLMs more factually correct and facilitating a more robust reasoning process. Knowledge graphs store big knowledge bases using graph-based data structures to store heavy corpus data in a structured format. KGs consist of nodes representing entities (such as places, people, objects) and edges representing relations (such as CapitalOf, BornIn, LocatedIn). KGs store facts in triple representation, i.e., head entity, relation, tail entity. This graph-based representation of relationships and entities, interlinking vast amounts of data from diversified sources, makes querying easier using SPARQL [6] and Cypher [7], enabling the updates to be done without continuously retraining the model, and help find plausibly correct responses. KGs improve factuality and interpretability through the graph-based retrieval and reduce hallucination, which stands out as the major problem of Large Language Models.

An organized summary of this survey is shown in Fig. 1, which also shows how LLMs and KGs complement one another. This survey shows a construction-first view of KG-LLM integration, as illustrated in Fig. 1. Although existing surveys treat KGs as given inputs to integration strategies, this survey argues that the KG construction paradigm, whether text-based, multimodal, or domain-specific, determines the structural properties of the resulting KG, which in turn governs the most appropriate integration strategy and the LLM tasks that become feasible. Fig. 1 shows the KG Construction Paradigms for the LLM Integration Framework proposed in this survey. The framework classifies the survey contributions into five layers. The bottom layer describes the data sources used as input for each construction paradigm — unstructured text and structured databases for general KG construction, multimodal data (e.g., images, audio, video) for multimodal KG construction, and electronic health records, clinical notes, and medical ontologies for

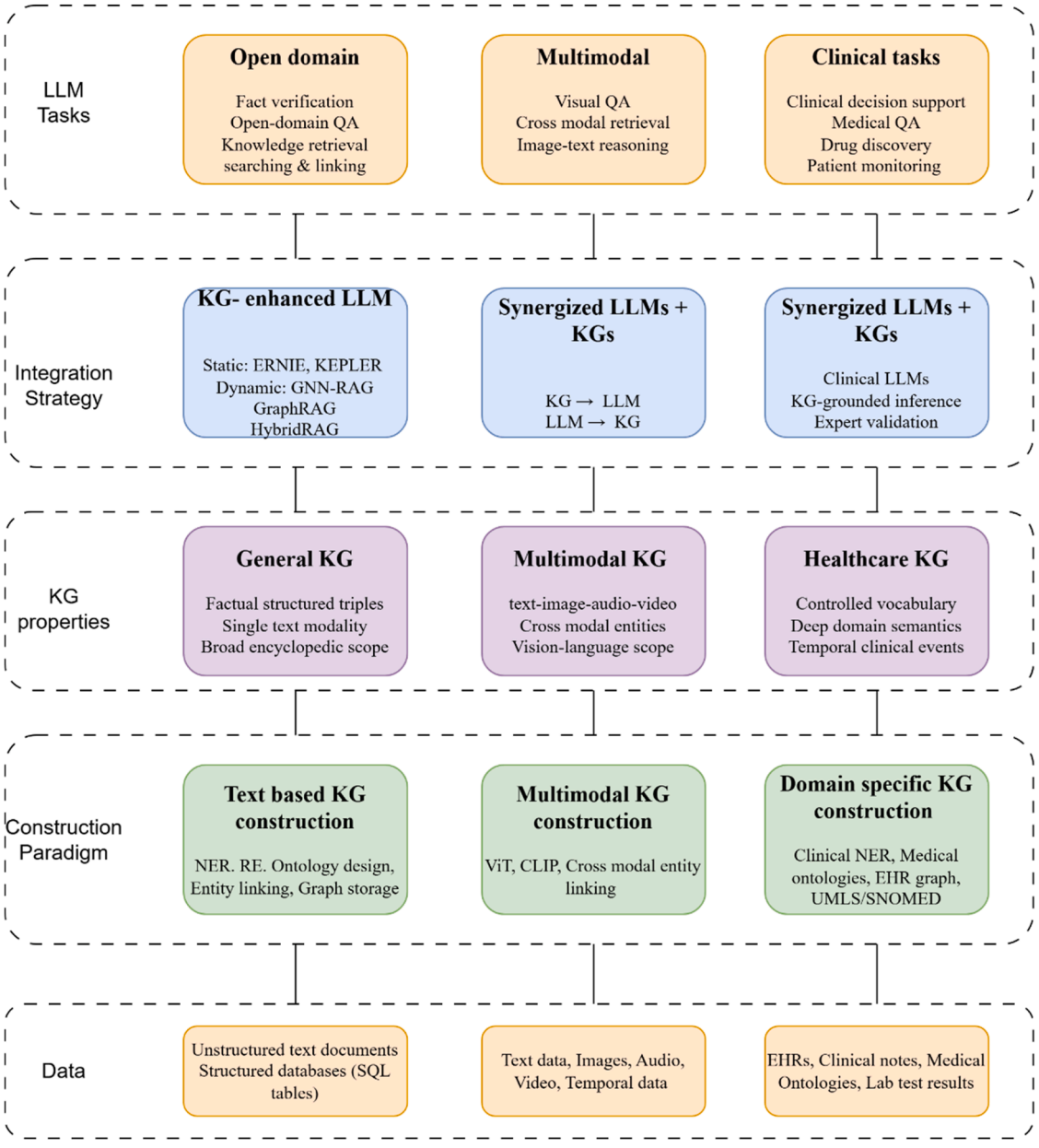


Fig. 1. Framework and Overview of KG-LLM Integration.

healthcare KG construction. The construction paradigm layer depicts the processing of each data type into a KG via different pipelines: NER, relation extraction, and ontology design for general KGs [Section 5], cross-modal entity linking and semantic fusion for multimodal KGs [Section 6], and clinical NER, medical ontology alignment, and EHR graph construction for healthcare KGs [Section 7]. The KG properties layer depicts how different construction paradigms yield structurally different KGs, specifically general KGs with factual structured triples, multimodal KGs with cross-modal entities, and healthcare KGs with controlled clinical vocabularies and temporal event structures. These structural properties directly determine the integration strategy layer. The general KGs are applicable to the KG-enhanced LLM approaches like GraphRAG [137] and HybridRAG [134], while the multimodal and healthcare KGs require the Synergized LLMs+KGs paradigm [131] for bidirectional enhancement. The top layer displays the LLM tasks enabled by each pathway: open-domain QA and fact verification for general KGs; visual QA and cross-modal retrieval for multimodal KGs; and clinical decision support and medical QA for healthcare KGs. Our survey differs from the existing works in that it considers the KG construction paradigm selection as the fundamental design choice that influences all downstream integration and task performance, rather than treating construction and integration as two separate problems.

This framework is realized by two analytical tables: Table 5 (Section 3.8) provides a systematic mapping of KG types to suitable integration tasks, analyzing which construction paradigm best suits each integration strategy; Table 12 (Section 8.4) extends Pan et al. [131] by comparing all strategies against applicable conditions and limitations – jointly forming an original decision framework for KG-LLM system design.

The latter part of this survey is structured as follows. Section 2 explains about the background of LLMs and KGs. Section 3 describes different type of KGs and the selection criteria. Section 4 presents how KGs mitigate LLMs failure modes. Section 5 showcases KG construction in detail. Section 6 shows construction of Multimodal Knowledge Graphs. Section 7 focuses on Domain-specific knowledge graph construction for healthcare data. Section 8 describes the Integration approaches and techniques for integrating KGs and LLMs. Section 9 entails different applications of KGs. Section 10 details on Evaluation Metrics used for KGs and Section 11 concludes the survey and outlines future directions and milestones to be achieved in this field of study.

2. Background

In this section, brief representation and architecture of LLMs will be introduced and then, we illustrate the deeper insights into the evolution of KGs in detail.

2.1. Large Language Models (LLMs)

Language Modeling (LM) has a significant impact on Natural Language Processing (NLP) for semantic understanding of text and response generation. Recent advancements of NLP in Artificial Intelligence (AI) have given rise to the Transformer architecture [8], which has been a breakthrough for LLMs such as Gemini 1.5/Gemini 2.0, LLaMA, and Mistral to generate human-like texts after being trained on billions of parameters. Recent ChatGPT-5 operates on text, image, audio, and video inputs, paving the way for multimodal integration. LLMs excel at generating responses that are grammatically nuanced to knowledge intensive. They are pre-trained on a large corpus by learning patterns from diversified sources for better generalization.

In the training process, they learn contextual relationships between words using an attention mechanism of Transformer [8]. Attention mechanism captures complex patterns and dependencies within data when performing predictions by emphasizing relevant parts of the input sequence [8]. This mechanism allows the model to do in-context learning to understand the full meaning of the input. Further improved, the self-attention mechanism enables us to weigh the importance of different parts of the input without considering their position in sequence. LLMs have proven to be so powerful and versatile that they excel as text summarization tools, to build chatbots, question & answering tasks, logical reasoning, and problem solving, making them next-gen intelligent systems. We now present different LLM model architectures:

2.1.1. Encoder-only LLMs

Models such as BERT enable bidirectional context to create detailed token representations, which makes them useful for tasks involving classification and retrieval. At every stage, they use self-attention layers to access all the words in the initial sentence. Often, the pre-training of these models masks the random words and tasks the model to predict the initial sentence. These models are best suited for tasks like question answering, sentence classification, named entity recognition [9]. Representatives of Encoder-only LLMs include BERT [10], ALBERT [11], RoBERTa [12], ELECTRA [13], DistilBERT [14] and are mainly used for classification, retrieval, and response generation.

2.1.2. Decoder-only LLMs

They use a decoder to generate the output text. At every stage, for a given word, attention layers access the words present before it in the sentence by processing the tokens sequentially from left to right. Often, the pre-training of these models is to predict the next word in the sentence, also referred to as Causal Language Modelling (CLM) [15]. These models are best suited for tasks like language generation, reasoning tasks, and multi-hop QA. Representatives of Decoder-only include many state-of-the-art LLMs like the GPT series. These models are fine-tuned based on LLaMA [16] and include Claude, Mistral, Falcon, and are used to perform downstream tasks when given simple instructions with not much of a prediction head added.

2.1.3. Encoder-decoder LLMs

They use both the encoder and decoder modules. The encoder is used to encode the input sentence by processing it bi-directionally and the decoder module is used to generate the output text. Often, the pre-training of these models is done by masking random words and replacing the masked word with single mask special token. The main task is to predict the token that this mask token replaces while preserving the meaning of the sentence. These are also referred as sequence-to-sequence models. These models are best suited for tasks like summarization, translation where in new sentences will be generated depending upon the given input. For example, T5 [17], BART [18] are good representatives of seq2seq (sequence-to-Sequence) models used to perform machine translation, text summarization, and question answering tasks.

2.1.4. Multimodal architecture

Multimodal LLMs integrate text, image, audio, and video data for comprehensive outputs, thereby enhancing creativity by generating innovative ideas and responses. These models address complex, dynamic, and knowledge-intensive challenges. Here, text encoder combined with visual/audio encoders are fed into a single transformer's shared embedding space, making it unified multimodal embeddings before reasoning. Some efficient representatives are GPT-4o [19], Gemini 2.0, Qwen2.5-VL, and VisionLLaMA, which are used in visual representation, image generation, and speech recognition.

Additionally, Table 1 provides a comprehensive summary of existing general-purpose and domain-specific LLMs, highlighting the steadfast expansion of LLMs in diverse application domains due to the availability of massive high-quality corpora and effective fine-tuning techniques that can adapt a model's knowledge to a particular domain requirement.

Table 1
Overview of General and Domain-specific LLMs.

Domain	Model & Year	Company	Parameters	Modality
General	GPT-3 [1], 2023	OpenAI	-	Multi modal
	GPT-4o [19], 2024		-	Text, Vision, Audio
	GPT-4 [22], 2023		-	Text + Vision
	Claude 3 Opus [3], 2024	Anthropic	-	Text, Vision
	Gemini 1.5 Pro [2], 2024	Google DeepMind	-	Multi modal
	LLaMA-3 [23], 2024	Meta	70B+	Text
	Mistral-7B [24], 2025	Mistral AI	-	Multi modal
	Qwen-2 [25], 2024	Alibaba	72B	Text
Healthcare	Med-PaLM 2 [26], 2023	Google	-	Text
	BioGPT [27], 2022	Microsoft	1.5B	Text
	ClinicalCamel [28], 2023	Open-source	13B	Text
Finance	BloombergGPT [29], 2023	Bloomberg	50B	Text
	FinBERT [30], 2019	Prosus AI	110M	Text
	FinMA [31], 2023	Shangai AI lab	~13B	Text
Scientific/ Research	SciBERT [32], 2019	Allen institute for AI	110M	Text
	Galactica [33], 2022	Meta AI	120B	Text
Code Generation	Code LLaMA [34], 2023	Meta	7B- 34B	Text
	StarCoder [35], 2023	BigCode	15B	Text
Autonomous Driving	DriveGPT4 [36], 2024	Research Team	7B+	Vision + Text
	DriveLM [37], 2023	NVIDIA	-	Vision + Text
Time- Series	LLMTime [38], 2023	Google research	-	Text
	Time-LLM [39], 2023	Shangai AI lab	7B	Text + Time series
	Chronus [40], 2024	Amazon AI	20M-710M	Time Series
Law	LawGPT [41], 2023	Research Team	~13B	Text
	LegalBERT [42], 2020	University of Oxford	~110M	Text
	Legal-LLaMA [43], 2023	Research Team	~7B-13B	Text
Recommendation	M6-Rec [44], 2022	Alibaba DAMO academy	~300M	Multi modal
	AgentCF [45]	Zhejiang University	~7B	Text

2.2. Knowledge graphs (KGs)

Knowledge graphs have existed since the 1980s on the semantic web and are the core to abundant information retrieval systems that have structured knowledge, be it domain-specific or domain-independent. The advent of Google's knowledge graph [20] in 2012 was the major turning point in the evolution of knowledge graphs, showcasing the shift from information engine to real-world knowledge engine through entity graphs embedded into AI systems, which enabled machines to understand the relationships between words. For instance, if the user searched for "Who is Albert Einstein?" the query result shows his picture, a short bio, his innovations, and even related people like his spouses, children, and his recognitions, enabling the computers to answer questions that align well. KGs store knowledge in graph-based data structures as a collection of triples, KG = {(h,r,t)⊆ E × R × E}, where E is the head entity, R is the relation, and E is the tail entity representing a set of entities and relations [21]. The nodes of the graph represent entities, like people, places, etc. The edges represent relations (like "is a scientist", "was born in") between different entities. KGs are used in semantic search to enhance search engines' semantic understanding and are also called semantic networks, as they store world knowledge about entities and their relations that are interrelated. Moreover, Table 2 showcases a summary of widely used general-purpose and domain-specific KGs enabling structured knowledge representation across various domains. Many large-scale KGs have been developed for different sectors like healthcare, finance, scholarly research, and enterprise management systems over the past few years. KGs excel at effective information retrieval, semantic reasoning and knowledge integration.

In the context of KGs, the term Resource Description Framework (RDF) [68] is a framework based on directed-edge labelled graphs,

Table 2
Overview of General and Domain-specific KGs.

Domain	Knowledge Graph & Year	Type	Scale
General	DBpedia [46], 2007	Encyclopedic KG	6M+ entities
	Wikidata [47], 2012	Collaborative KG	100M+ entities
	YAGO [48], 2007	Semantic KG	10M+ entities
	Freebase [49], 2007	General KG	50M+ entities
Healthcare	SNOMED CT [50], 2002	Clinical Ontology	350K+ medical concepts
	UMLS [51], 1990	Biomedical meta-thesaurus	4M+ concepts
	DrugBank KG [52], 2006	Drug knowledge graph	14K+ drugs
Commonsense	ConceptNet [53], 2008	Commonsense KG	34M+ edges
	ATOMIC [54], 2018	Inferential commonsense KG	877K+ triples
	Comet [55], 2019	Inferential commonsense KG	~1.3M+ entities
Enterprise	Google KG [20], 2015	Web-scale KG	Billions of entities
	Microsoft Satori [56], ~2013	Search KG	Large-scale
Finance	FIBO [57], 2013	Financial Ontology	100K+ concepts
Legal	LKIF-core [58], 2008	Legal ontology	Structured
Scholarly Research	Microsoft Academic Graph (MAG) [59], 2015	Scholarly citation KG	200M+ papers
	Open Research KG (ORKG) [60], 2018	Scholarly KG	54K+ papers
Cybersecurity	OpenCTI [61], 2018	Threat intelligence KG	6M+ entities
Recommendation	Amazon product KG [62], ~2015	Product KG	70B+ triples
Autonomous Driving	nuScenes KG [63], 2019	Autonomous driving sense KG	1.4M+ annotated objects
Multimodal	IMGpedia [64], 2017	Image KG	~14.7 M images
	MMKG (Visual Genome KG) [65], 2016	Multimodal KG	108K images
Time-series	GDELT KG [66], 2013	Global event KG	Trillions of datapoint and billions of entities
	ICEWS [67], ~2009	Event-based KG	20M+ events

which has been recommended by the World Wide Web Consortium (W3C) that provides semantic richness and standardization for representing information on the web. It specifies data in triple representation. A triple consists of a subject (a resource), an object (a resource) and a predicate that indicates the relationship between them. Resources can be any entity, like people, places, documents, etc. RDF denotes the subject, object and predicate using IRIs (Internationalized Resource Identifiers) and is also called its referent. Many open knowledge graphs like DBpedia [46], which is a large-scale, structured knowledge base extracted from Wikipedia, YAGO [48], a core of semantic knowledge, Wikidata [47], a free collaborative knowledge base, adopt the RDF framework and are useful to build KGs that incorporate the relationships between multiple resources on the web.

Moreover, Table 3 compares this study with the existing surveys [9,21,80,93,110–113,139,145,131,132,138,108] including LLM Architecture (Encoder / Decoder Models), Fine tuning mechanisms, Hallucination and Failure Analysis. The surveys encompass different KG types and give an overview of KG construction and Multimodal KG construction. They cover LLM and KG integration strategies, discuss RAG, GraphRAG, and domain-specific KGs in healthcare, and present the corresponding evaluation metrics.

Many existing surveys focus on Information extraction of the KG construction pipeline. A comprehensive analysis of deep learning techniques for named entity recognition (NER) is provided by Li et al. [9], who analyzed the evolution of transformer-based encoders like BERT and RoBERTa from recurrent sequence labeling models. This scope is expanded to open information extraction (OIE) by Pai et al. [93], exploring the techniques that generalize relation extraction and NER to open-domain settings without preset schemas. Although the extraction layer is well covered in both works, retrieval-augmented generation pipelines, LLM integration, and downstream KG construction are not addressed.

Meanwhile, the researchers have investigated the foundational vocabulary and taxonomy of knowledge graphs. One of the most authoritative treatments of KGs is provided by Hogan et al. [21], establishing the canonical taxonomy of ontological, encyclopedic, and commonsense KGs covering formalisms, construction methods, storage, querying, and embedding-based reasoning. Additionally, Zou [145] provides a survey of KG applications in the domains of recommendation, question answering, and healthcare. Abu-Salih [80] further explores domain-specific KGs with a focus on biomedical and clinical graph construction. Nevertheless, none of the three surveys stated above discuss multimodal KG paradigms or graph-based retrieval, and they all either predate or do not use large language models as construction and reasoning tools.

Specialized sub-problems within the KG lifecycle are addressed by another set of surveys. Zhu et al. [110] review embedding-based and iterative techniques for reconciling equivalent entities across heterogeneous KGs, with a sole focus on knowledge graph entity alignment. Neuro-symbolic reasoning over KGs is studied by DeLong et al. [111], and survey techniques that integrate symbolic logic and neural representation learning for multi-hop inference. None of these works offers a comprehensive approach to the entire KG construction pipeline or its integration with large language models, even though each one delves deeply into its sub-problem. Chen et al. [113] survey multimodal knowledge graph construction methods. Cross-modal alignment challenges unique to MMKG-LLM integration, such as the modality gap [149], missing modality scenarios [90,91,122], and noisy alignment propagation [64,115] are not covered, constituting a gap that this survey explicitly fills via dedicated analysis in Section 6.9.

The integration of knowledge graphs with large language models is the most closely related category of surveys. In their influential roadmap, Pan et al. [131] discuss hallucination mitigation as a cross-cutting issue and suggest the taxonomy of KG-enhanced LLMs, LLM-augmented KGs, and synergized LLMs and KGs. LLM-driven KG augmentation techniques are surveyed by Ibrahim et al. [132], covering relation discovery, entity typing, and completion. Zhu et al. [112], examine LLMs for KG reasoning and construction, including entity extraction, relation extraction, and KG-based question answering. This view is extended to LLMs on graph-structured data in general by Chen et al. [138], focusing on prompting and fine-tuning techniques, including parameter-efficient techniques like LoRA. The surveys stated above represent LLM-KG integration, but multimodal KG construction and healthcare-specific clinical KG construction are not addressed.

Peng [108] examines graph retrieval-augmented generation (Graph-RAG), covering both indexing strategies and hybrid dense-sparse retrieval architectures. Evaluation metrics like precision, recall, F1, MRR, and nDCG are rigorously analyzed across classification and ranking tasks by Rainio et al. [139] in their systematic review of evaluation metrics and statistical tests in machine learning. Although both surveys offer useful methodological foundations, neither discusses domain-specific KG deployment in clinical settings nor the entire KG construction pipeline.

This survey is different from the existing surveys in the following aspects:

Table 3
Comparison of This Work with Existing Surveys.

Ref.	LLM Background		Knowledge Graph Coverage				LLM-KG Integration Strategies			RAG	Domain and Evaluation	
	LLM Architecture (Encoder/ Decoder)	Hallucination & Failure Analysis	KG Types & Taxonomy	KG Construction pipeline (NER→RE)	Multimodal KG construction	Cross-modal Alignment Challenges	KG-enhanced LLMs	LLM augmented KGs	Synergized LLMs & KGs (Neuro-symbolic)	GraphRAG & HybridRAG	Healthcare & Clinical KG	KG Evaluation Metrics
[9]	√											
[21]			√	√								
[80]			√	√							√	
[93]				√								
[108]								√	√	√		√
[110]			√									
[111]									√			
[112]	√		√	√			√	√				√
[113]			√	√	√	√						
[131]	√	√	√				√	√	√			√
[132]	√		√				√	√	√			√
[138]	√		√					√				
[139]												√
[145]			√								√	
Our Work	√	√	√	√	√	√	√	√	√	√	√	√

1) In terms of KG construction coverage, we offer a thorough end-to-end treatment of the KG construction pipeline, integrating both traditional techniques and the most recent LLM-based approaches [9,93]. This includes named entity recognition and relation extraction through entity linking, ontology alignment, and graph storage. The entire pipeline in the context of LLM is not covered by current surveys, which either focus solely on the extraction layer [9,93], or the KG product itself [21,80]. We provide a benchmark comparison of NER and ontology alignment tools with empirical performance data [98,102] and tool selection guidance, which is lacking in most of the surveys.
2) In terms of multimodal knowledge graphs, we offer specialized coverage of MMKG construction techniques that combine textual, visual, and structured modalities using LLMs as cross-modal encoders. Specifically, this survey tackles cross-modal alignment issues like modality gap [149], missing modalities [90,91,122], and noisy alignment [64,115], which are not discussed in the dedicated MMKG survey of Chen et al. [113].
3) In terms of LLM-KG integration strategies, this survey extends the work of Pan et al. [131], by explicitly covering GraphRAG and HybridRAG as practical retrieval implementations of the synergized paradigm [108] illustrating a coverage gap in Pan et al. [131] and all other compared surveys. An original analytical framework has been proposed for mapping types of KGs to appropriate integration strategies, comparing all integration approaches across core mechanisms, applicable conditions and limitations [111,152]. Neuro-symbolic reasoning is demonstrated as the fundamental paradigm for KG-LLM integration [111,152].
4) In terms of Domain Applications, we provide a focused discussion on healthcare and clinical knowledge graphs, including medical ontologies, clinical NLP pipelines, and LLM-driven biomedical KG construction [80,145]. This survey goes beyond existing domain surveys by explicitly discussing four healthcare-specific construction challenges, including mandatory de-identification [94,129], temporal EHR reasoning [129,130], multi-ontology alignment across ICD-10, SNOMED-CT, and UMLS [50,51,124], and data sparsity of rare diseases [52], via a dedicated comparative analysis of general versus healthcare KG construction methodologies.
5) In terms of evaluation, this survey reviews a range of KG-specific evaluation metrics, including entity and relation extraction F1 scores, link prediction metrics such as MRR and Hits@k, KG completion benchmarks, and hallucination detection measures. We organize these metrics within a unified evaluation framework that links the quality of KG construction to the downstream performance of LLM-based systems [139]. Standard KGQA benchmarks, such as WebQuestionsSP [151] and KQA Pro [150], are considered in evaluating KG-LLM integration strategies [139].
6) In terms of future directions, this survey presents a prioritized research agenda for five directions, including 1) dynamic KG updates, 2) cross-modal alignment, 3) reasoning over incomplete graphs, 4) standardized end-to-end benchmarks, and 5) ethical bias propagation between KGs and LLMs, along with concrete research questions and proposed solutions for each direction.

In summary, to the best of our knowledge, this survey provides a comprehensive and unified overview that adopts an integrative construction paradigm bringing together the entire KG construction pipeline, multimodal KG paradigms, all three major LLM–KG integration strategies, GraphRAG and HybridRAG approaches, healthcare-focused KG applications, and systematic evaluation methodologies within a single work.

3. Types of knowledge graphs

Different types of KGs are discussed below:

3.1. Property graphs

They are widely used in eminent graph databases, such as Neo4j [69] to add flexibility when modeling data as a graph. This allows attributes to be added to edges and nodes and provides fast graph traversal without creating additional triples. Here, the edges can have multiple attributes, making them handle multi-faceted relations, and properties can differ for every node. RDF graphs often use triple structure, whereas property graphs use multi-labeled nodes and edges, featuring their properties as key-value pairs. S3PG is a technique that can convert RDF KGs to property graphs [70]. Fig. 2 depicts a scenario of building an

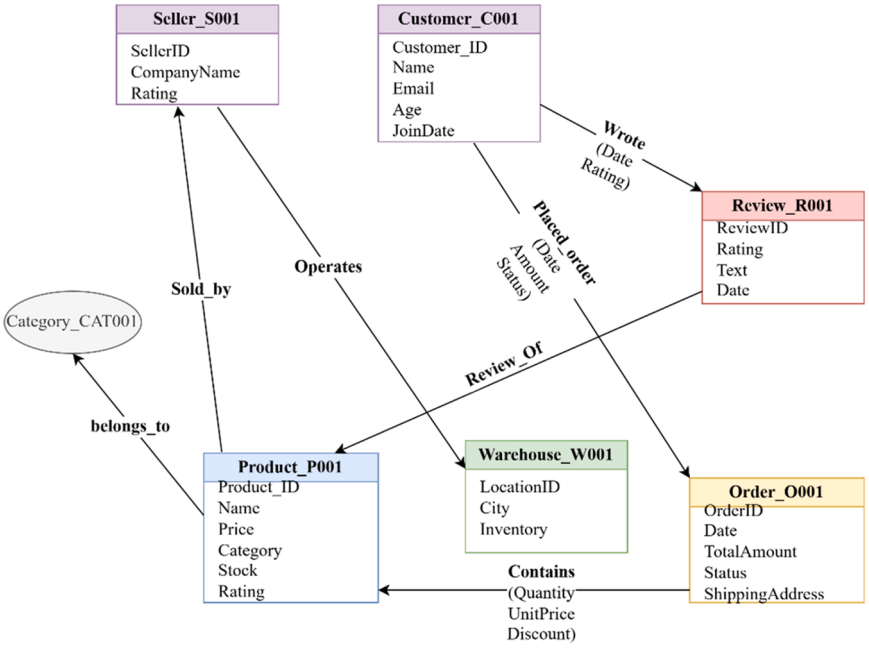


Fig. 2. Property graph model of an e-commerce system with enriched property nodes and edges.

e-commerce system for an online marketplace where every node is associated with its properties that allow fast querying and traversal.

3.2. Hypergraphs

Mainly, knowledge graphs are represented by binary relations (an edge connecting two nodes), but when a single edge (called a hyperedge) links multiple nodes is said to form a hypergraph [21]. This is the combination of a set of vertices and hyperedges that connects multiple nodes with a single edge [71]. This is used in modeling complex relationships where a single edge can connect a tuple of entities forming n-ary relations. A hypergraph can connect multiple nodes, that represents multi-dimensional relationships and connections between the nodes [72]. Fig. 3 depicts an example where a project in a particular year can be completed by the collaboration of multiple researchers of an institution, forming a scientific event. Hypergraphs represent this without having it to break it into multiple triples. The hyperedges ResearchProject, FundedResearch, and Publication have properties associated, that help in a better understanding of the relationships. These graphs are good at representing group-level interactions and higher-order relationships. YAGO [48] ontology represented a way to use n-ary facts like "bornOnDate(person, date, place)" for representing relationships as entities.

3.3. Enterprise knowledge graphs

A knowledge graph built for a particular company to organize and unify its data from disparate silos by merging them into a single graph of entities and relationships. Data from heterogeneous sources like databases, spreadsheets, and documents can be structured to form a graph schema. As the resulting graph would be very large, scaling from millions to billions of nodes/edges, refinement steps are applied to clean and refine the KG. Georgia Institute of Technology has built one such enterprise knowledge graph to make a structured representation of its researchers' collaboration efforts within and outside the institution [73]. They have built a KG on publications by GT researchers using Neo4j, which consists of 471,396 nodes and 2,155,302 relationships while the KG was in production. Many big tech companies have their own enterprise KG, namely, Microsoft [74], Airbnb [75], Amazon [76], Uber [77], and LinkedIn [78], to improve knowledge discovery within a company. Recent research shows how integrating structured, semi-structured, and unstructured data

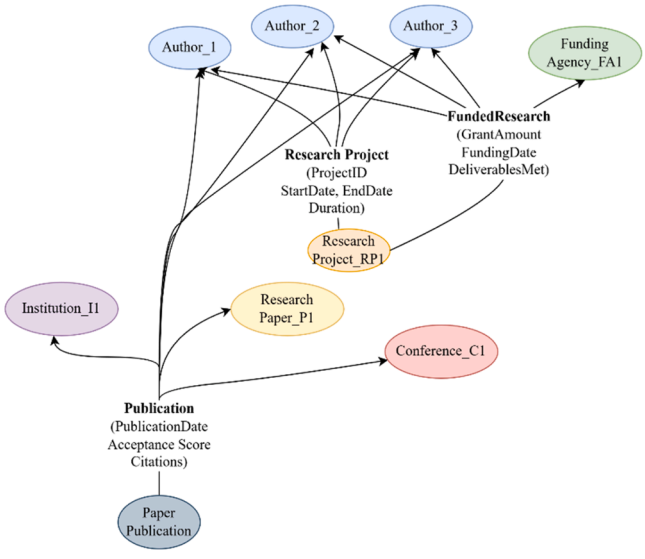


Fig. 3. Hypergraph model for research ecosystem linking multiple researchers and publications through one hyperedge.

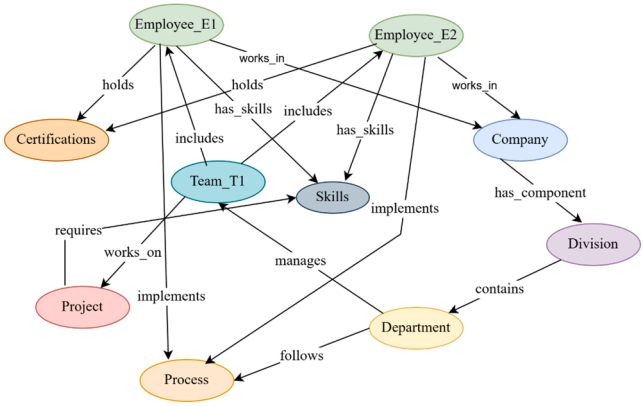


Fig. 4. Enterprise KG showing organizational hierarchy across company, division, department, team, and employee nodes.

Table 4
Management of Organizational Hierarchy.

Layer	Entity	Properties
Organization	Company	CompanyId, Name, FoundedYear, Headquarters, Revenue, EmployeeCount
	Division	DivisionId, Name, ParentCompany, Budget, HeadCount
	Department	DeptId, Name, ParentDivision, Budget, Responsibility
	Team	TeamId, Name, ParentDept, Focus, TeamSize
Personnel	Employee	EmployeeId, Name, Title, Department, Salary, StartDate, Manager, Skills, Certifications
	Contractor	ContractorId, Name, Skills, CompanyContract, RatePerHour
Capabilities	Skills	SkillId, SkillName, Category, ProficiencyLevels
	Certifications	CertId, CertName, IssuingBody, ExpiryDate, Renewal
	Competencies	CompId, Name, RequiredSkills, Level
Project	Project	ProjectId, Name, StartDate, EndDate, Budget, Status, Owner
	Process	ProcessId, Name, Steps, Owner, Documentation, Version
	Initiatives	InitId, Name, Goal, Timeline, Leadership

sources can result in a strong knowledge base (KG) that enhances analytical capabilities in an enterprise setting and facilitates downstream knowledge discovery [79]. Fig. 4 illustrates an enterprise knowledge graph showcasing a scenario for managing organizational hierarchy, teams, and the capabilities of employees in a company. Table 4 summarizes the management of organizational hierarchy in an enterprise KG shown in Fig. 4.

3.4. Encyclopedic knowledge graphs (Search engine KGs)

Search Engine KGs are used to perform web searches and enhance the results of queries by using their structured knowledge. It also aids with query understanding, by looking at the KG context, it understands that jaguar can mean an animal or a car brand. Encyclopedic KGs are created by ingesting data from diversified sources like Wikipedia and public databases. Fig. 5 depicts an encyclopedic knowledge graph that shows the demographic details, inventions and the research interests of Nobel laureates. Wikidata [47] is one such encyclopedic knowledge graph, which has a diverse knowledge base drawn from Wikipedia. YAGO [48] also has a massive knowledge repository of semantic understandings. Freebase [49] is a database used to structure human knowledge, which has public read/write access, and DBpedia [46] is also derived from Wikipedia.

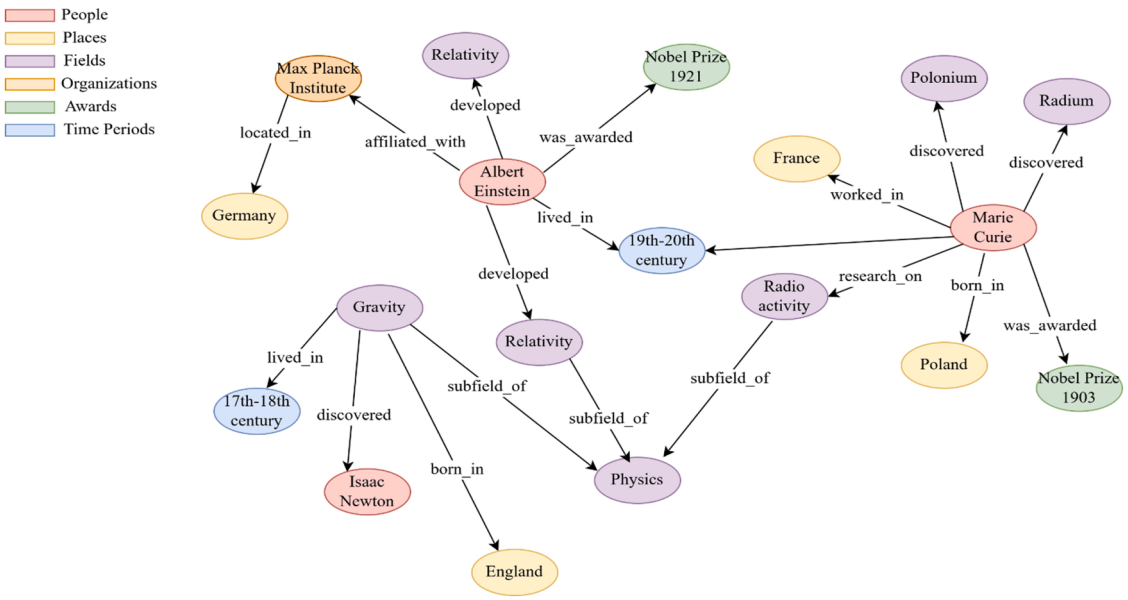


Fig. 5. Encyclopedic KG graph of Nobel laureates with organizations of demographics, inventions and their research interests.

3.5. Domain-specific knowledge graphs

These KGs have unique schemas and databases that are specific to a field, which contain a profound understanding of that area and are designed to ease the response generation to support expert queries. Domain-specific KGs are often more reliable and represent a new form of data analytics that depicts semantically interconnected datasets [80]. Fig. 6 illustrates a health-specific KG that shows people associated with different diseases, doctor treatments, prescribed medications, and the smart device they use to monitor health-specific parameters, which gives a complete scenario of a person and their underlying condition. Unified Medical Language System (UMLS) [51] is a medical domain KG framework that connects synonyms, defines relationships, and integrates various medical terminologies to enhance interoperability among healthcare applications. In addition to this, Bio2RDF [81] builds mashups of bioinformatics data. Some other domain knowledge graphs to name a few, include, Finance [82], Legal [83], and Chemistry [84].

3.6. Common-sense knowledge graphs

Common sense KGs exhibit general world and experiential knowledge that includes basic human commonsense about daily concepts and relationships [85]. These graphs help computers understand the meaning of sentences that humans generally use. For example, the text is extracted as (Ball, usedFor,Play) for a sentence like, Ball is used for playing. Fig. 7 depicts an illness, fever, that is connected to its causes, related symptoms, medications and the device used to detect the rising temperature of a human body. ConceptNet [53] is a multi-lingual KG that is useful in language understanding by connecting words with labeled edges like bornAt, capableOf, atLocation. Its knowledge is extracted from WordNet (A database of semantic relationships) and WikiHow, crowd sourcing, and games with a purpose [53]. Some other

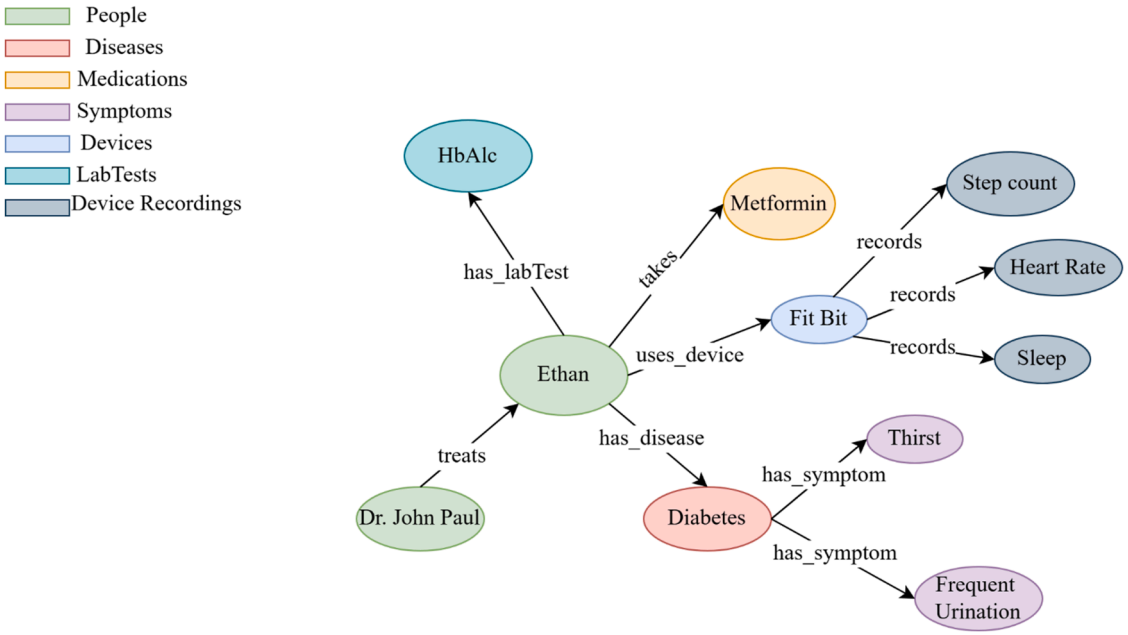


Fig. 6. Domain-specific healthcare KG linking patients, diseases, treatments, and monitoring devices.

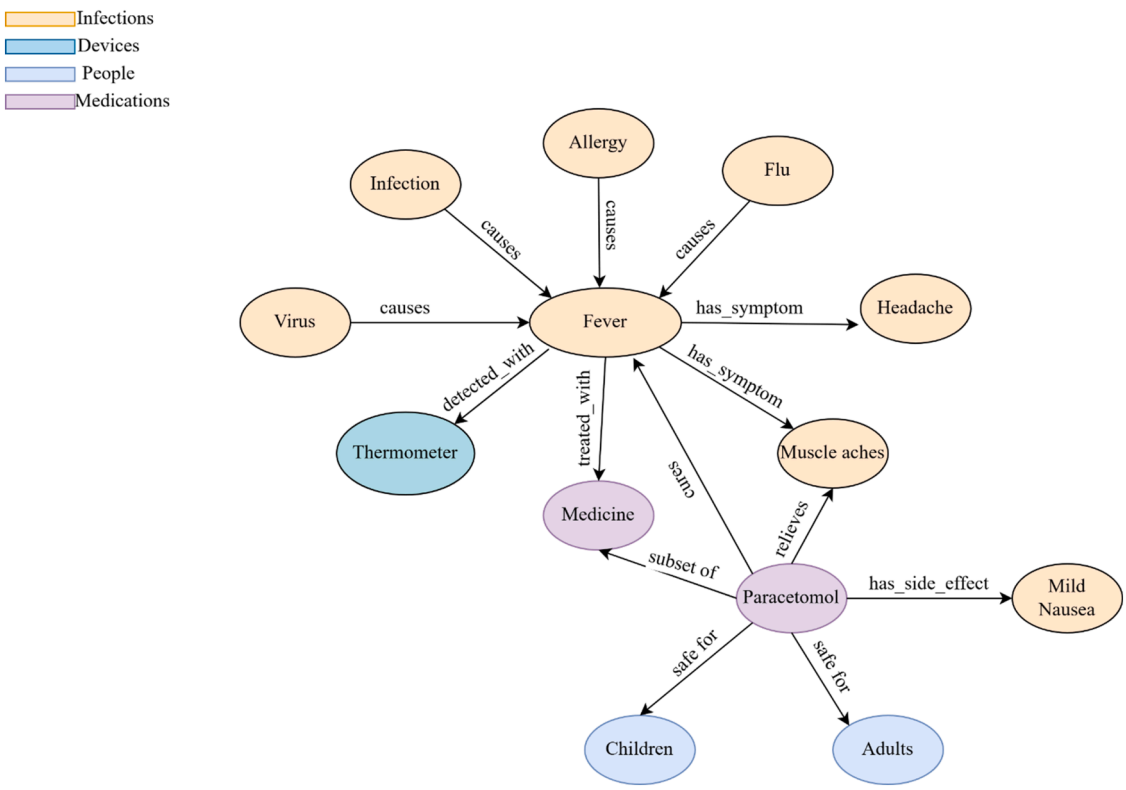


Fig. 7. Common-sense KG model for representing everyday health reasoning.

common-sense knowledge graphs, such as Atomic [54], focus on knowledge classified as if-then relations with variables, whereas Comet [55] has everyday inferential knowledge tuples about entities and events. The Cyc knowledge base [54] is widely used for building a large-scale commonsense reasoning system.

Besides the well-known ATOMIC and COMET-ATOMIC 2020, recent works have opened new possibilities for commonsense KGs. TransOMCS [86] automatically extracts commonsense knowledge from linguistic graphs extracted from web text sources, including Wikipedia, Yelp, and Reddit, producing 18.48 million tuples using ConceptNet relations – offering significantly larger scale at the cost of lower precision compared to manually curated graphs. More recently, ECoK [87] has been proposed to fill an important gap among existing commonsense knowledge graphs by focusing specifically on emotional knowledge. ECoK builds rich emotional commonsense knowledge that is not well addressed by existing general-purpose KGs by combining theories from psychology, cognitive science, and linguistics with LLM-based construction techniques.

3.7. Multimodal knowledge graphs

Traditional KGs are grounded only to textual data, but Multimodal KGs integrate semantic and heterogeneous information, which includes image, audio, video, sensor, and give them a graph structure. Fig. 8 shows the combination of entities, techniques, applications, and temporal data that uses multimedia resources such as images, audio, video, and documents that are interconnected through semantic relationships. For example, MMKG [65] is a benchmark dataset that uses a collection of three knowledge graphs constructed from Freebase, DBpedia, and YAGO, where every entity is enriched with relationship triples, has numerical attributes, and an associated image with precomputed image embeddings. This dataset enables research on multimodal link prediction and entity matching tasks using cross-graph entity alignments.

On the other hand, Multimodal knowledge graphs have become more specialized and more comprehensive in recent years. For example, VisualSem [88] is a high-quality multimodal KG for vision-language tasks, connecting semantic concepts from BabelNet to over 930K images for 89,896 entities. Unlike general-purpose MMKGs, VisualSem emphasizes semantic richness and image quality, thus being especially suitable for tasks such as image captioning, visual question answering, and cross-modal retrieval. More recently, UKnow [89] proposes a unified knowledge protocol integrating multimodal knowledge across text and visual modalities, spanning 1.4M entities and 1.1M relations. UKnow solves a major limitation of existing MMKGs by proposing a unified framework for multimodal knowledge representation that supports both reasoning and vision-language pre-training tasks.

Richpedia [90] provides a pathway for KG and computer vision tasks, such as link prediction and visual relation detection. Some other examples include TIVA-KG [91], ImageGraph [92] and IMGpedia [64].

3.8. KG types for LLM tasks

The suitability of each KG type for particular LLM integration tasks is directly determined by its structural properties, as summarized in Table 5. Property graphs [69,70] can help LLM-driven query generation tasks, such as natural language to Cypher translation. Hypergraphs [71,72] extend this to multi-relational medical reasoning, where group relationships among multiple entities are to be captured at the same time. Extensive factual coverage makes encyclopedic KGs [46,47] widely adopted for open-domain QA and fact verification, but knowledge staleness hinders their reliability for time-sensitive queries. Domain-specific KGs [51,129,130] address clinical decision support tasks, where controlled medical vocabularies greatly reduce LLM hallucination. Commonsense KGs [53–55] encode implicit if-then event relations critical for dialogue generation and reasoning tasks, with recent extensions including emotionally grounded knowledge in ECoK [87] and automatically extracted commonsense in TransOMCS [86]. Multimodal KGs [65,88,89] could realize cross-modal grounding for visual QA tasks, which is impossible for purely textual KGs. A consistent pattern across all types is that the more specialized the KG structure is,

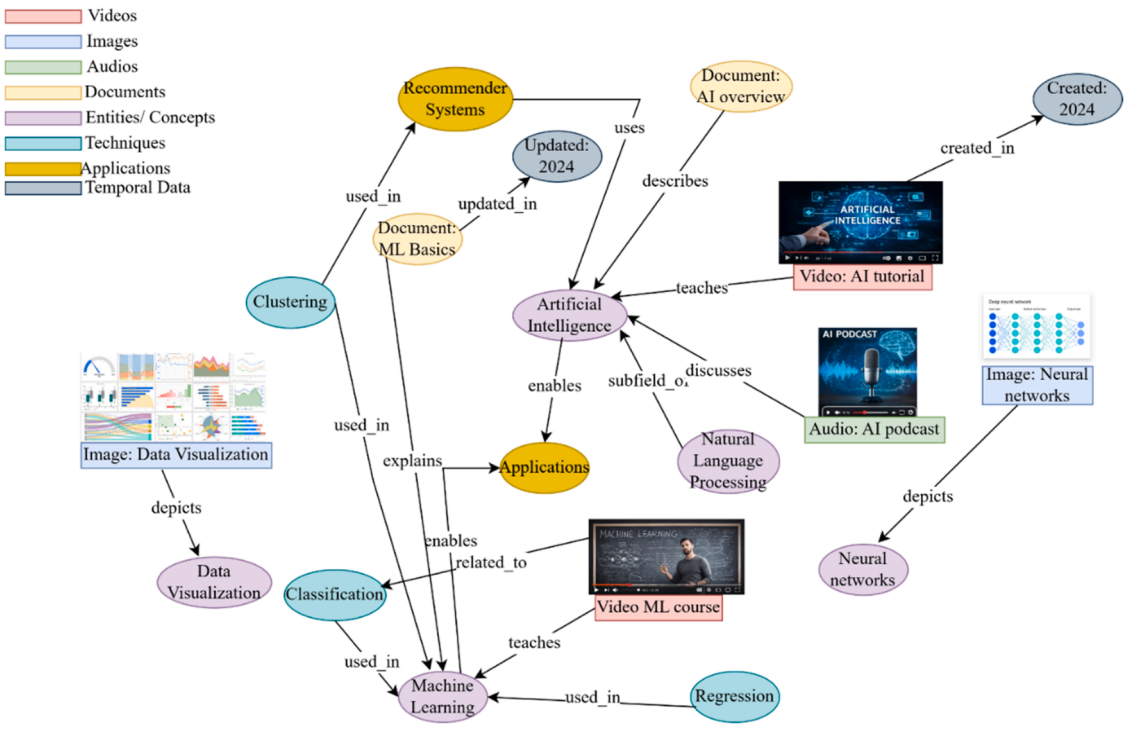


Fig. 8. Multimodal KG representing cross-modal entities.

Table 5
Knowledge Graph Types for LLM Integration Tasks.

KG Type	Commonly Applied LLM Task	Rationale	Key Limitation
Property Graph [69,70,73]	Knowledge graph question answering, Entity relationship extraction	Flexible node-edge-property structure enables rich relational queries and LLM-driven Cypher generation	Schema heterogeneity across domains complicates cross-graph generalization
Hypergraph [71,72]	Multi-relational reasoning, Complex medical knowledge representation	Hyperedges capture group relationships among multiple entities simultaneously — beyond pairwise triples	High computational complexity; limited LLM tool support for hypergraph traversal
Enterprise KG [62,79,80]	Knowledge discovery, Personalized recommendation, CRM	Business entity relationships and organizational hierarchies support context-aware LLM inference	Proprietary data restricts sharing; limited generalizability beyond organizational context
Encyclopedic KG [21,46–48]	Open-domain QA, Fact verification, Entity linking	Broad factual coverage and structured triples provide reliable grounding for general LLM inference	Knowledge staleness; limited reasoning depth for multi-hop queries
Domain-specific KG [51,80,129,130]	Clinical decision support, Medical QA, Drug discovery	Controlled vocabulary and deep domain semantics reduce LLM hallucination in specialized tasks	High construction cost; narrow applicability outside the target domain
Commonsense KG [53–55,85–87]	Commonsense reasoning, Dialogue generation, Emotion recognition	If-then event relations capture implicit everyday knowledge absent from encyclopedic KGs	Sparse coverage; noisy automatic construction; cultural bias
Multimodal KG [65,88–91,113]	Visual QA, Cross-modal entity linking, Multimodal retrieval	Aligned text-image-audio entities enable cross-modal grounding for vision-language LLM tasks	Cross-modal alignment noise; high construction and annotation cost

the more targeted are its applicable LLM tasks. This suggests that KG type selection should be a task-driven design choice, not a generic infrastructure choice.

4. How KGs mitigate LLM failure modes

Though the LLMs are powerful, they face challenges such as factual hallucinations, outdated knowledge, as they have information only as up to date as the data they were trained on and must rely on external retrieval processes and lack domain-specific expertise. Thus, there is a need to unify LLMs and KGs and enhance their performance. The following factors can be achieved through this unification:

4.1. Dynamic real-world knowledge management

LLMs are black-box models that store information from diversified databases in fixed parameter weights after training. Data on the web frequently keeps changing, like people, organizations, presidents, CEOs, etc. This requires updating the corpus and retraining the model, which is an expensive process.

Meanwhile, KGs help isolate facts from language models. When a particular entity relationship changes, for example, a new president, only the very graph needs to be updated, not the entire corpus of the LLM. This helps in less expensive knowledge updates and ensures access to real-time and factual data without retraining.

4.2. Reduce hallucinations

LLMs often generate plausible but incorrect information as they rely on their corpus, which has probabilistic correlations rather than grounded facts. It predicts the next word based on patterns learned from huge corpora. KGs have verified truths that are used by LLMs in response generation, making the generation pipeline more grounded using real-world triples, reducing the risk of hallucinations.

4.3. LLMs become more interpretable

LLMs being black-box models, the reasoning is usually hard to explain why a certain answer is generated. KGs have structured representations of facts where every inference path can be seen, for example, Disease→treatedBy→Drug. KGs follow an explainable chain of reasoning and when symbolic KGs combine with sub-symbolic LLMs result in neuro-symbolic interpretability that helps to check outputs before final generation.

4.4. Cost and efficiency

Retraining an LLM to update it with new knowledge is computationally expensive and internal reparameterization is time-consuming at almost all times. KGs can be updated dynamically and do not need to retrain the model for updates done on corpora. This eventually helps in reducing latency and cost.

5. Knowledge graph construction

As the previous section highlights the functional role of KGs in mitigating LLM failure modes, we now discuss the methodologies for constructing knowledge graphs. The construction follows a multi-stage pipeline comprising the key technologies such as building a data-driven ontology model, knowledge extraction, knowledge fusion (refinement & completion), and knowledge reasoning. Prior to this multi-stage pipeline, we have a data layer that gathers input data from diversified sources. Fig. 9 illustrates the detailed representation of an architecture of knowledge graph construction explaining the process step by step in an insightful manner. Table 6 lists different tools and techniques at every phase of KG construction, from data collection to knowledge retrieval and gives a clear overview of examples highlighting the practical use of tools. Table 7 is a benchmark performance comparison table that showcases empirical benchmark results for the NER and ontology alignment tools.

The multi-stage pipeline is as follows:

5.1. Data acquisition layer

Input data is gathered here from various sources, and the data is heterogeneous, such as PDFs, logs, Images, CSVs, multiple databases, etc. This layer consists of both structured data, like relational databases, APIs, and unstructured data like reports, texts.

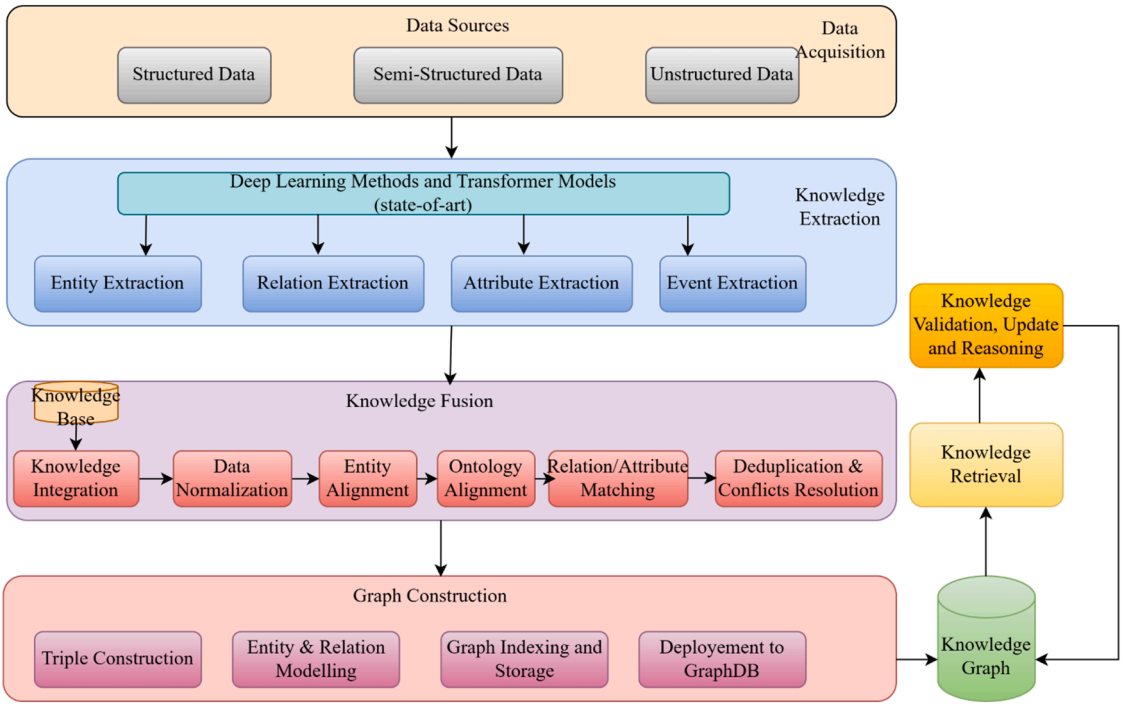


Fig. 9. Multi-stage Knowledge Graph construction pipeline.

Table 6
Tools used in each phase of KG Construction.

Phase	Tools/Techniques	Examples
Data Collection	APIs, Raw data from sensors	UMLS API [51], FHIR API [94]
Data Preprocessing	Normalization tools	NLTK cleaning [95], SpaCy preprocessors [96]
Knowledge Extraction	Named Entity Recognition (NER), Named Entity Linking (NEL), Named Entity Disambiguation (NED), Relation Extraction, OpenIE	Stanford NER [97], OpenNLP [98], OpenIE [93], SpaCy NER [99], BERT-NER [10], FOX [100], Open Calais [101]
Knowledge Fusion	Ontology mapping tools	LogMap [102], COMA++ [103]
Graph Construction	RDF/OWL conversion tools	Neo4j ETL(Extract-Transform-Load) [104], RDFLib [105], Jena [106]
Indexing & Storage (Graph DB)	Graph DB, indexing engines	Neo4j [69], JanusGraph [21], PostgreSQL [107]
Knowledge Retrieval	Query languages, APIs, graph retrieval tools	SPARQL [6], Cypher [7], Neo4j Browser [104]
Application Layer / LLM Integration	LLM + KG unified pipelines	GraphRAG [108], FAISS (Facebook AI Similarity Search) [109]

Table 7
Benchmark performance comparison of NER and ontology alignment tools used in KG construction pipelines.

Tool	Task	Benchmark	Precision (%)	Recall (%)	F1 (%)	Ref
StanfordNLP	General NER	CoNLL-2003	90.06	73.67	81.05	[98]
SpaCy	General NER	CoNLL-2003	60.94	49.01	54.33	[98]
Gate	General NER	CoNLL-2003	61.48	47.44	53.55	[98]
OpenNLP	General NER	CoNLL-2003	68.68	30.44	42.18	[98]
NLTK	General NER	CoNLL-2003	51.78	45.56	48.47	[98]
BERT-BiLSTM-IDCNN-CRF	Domain-specific NER	Chinese Geological NER	89.00	87.00	88.00	[126]
LogMap	Biomedical ontology alignment	FMA-NCI (OAEI)	81.10	84.00	82.50	[102]
LogMap	Biomedical ontology alignment	Mouse-NCI Anatomy (OAEI)	91.80	82.60	87.00	[102]
COMA++	Schema/ontology matching	Large real-world schemas	High quality reported	-	-	[103]

5.2. Building a data-driven ontology model

In the semantic web, Ontology refers to the machine-readable representation of knowledge that provides a formal understanding of data by defining entities(classes), instances, relationships, properties, and rules that connect all of these. For example, entities(classes) can be a Patient, HeartRate, and an instance is a patient name(patientXYZ), relationships and properties refer to a particular entity, like patientXYZ hasHeartRateValue, and rules refer to the constraints imposed, like heart rate must be a numerical value. Ontologies make data interoperable between heterogeneous datasets by having a shared vocabulary and can be represented using RDF or OWL and a property graph.

5.3. Knowledge extraction

The goal of this step is to identify entities, relations, and attributes from data. The advanced techniques such as Named Entity Recognition (NER), are used to find entities in the text, Named Entity Disambiguation (NED) is used to find the exact meaning a particular entity refers to, and Named Entity Linking (NEL) is used to link the entities to knowledge graph IDs. In addition to entity extraction and entity linking, relation extraction is performed using rule-based approaches like dependency parsing, pattern matching, or using supervised machine learning algorithms like CNN, LSTM, and logistic regression that use labeled training data. The evolution also involves transformer-based pre-trained large language models to derive structured information from unstructured text that has been summarized with a chronological framework representing OpenIE [93]. Attribute extraction is an important step that identifies properties of entities and event extraction retrieves temporal events from the text. The extraction process involves deep learning approaches like RNN/LSTM/BiLSTM-CRF models, Transformer-based models (BERT), and CNN-based NER, which provide better generalization in different domains and perform good semantic reasoning.

Benchmark comparisons give empirical guidance for selecting tools at each stage of KG construction. For general-purpose NER, Schmitt et al. [98] report that StanfordNLP achieves the highest overall F1 of 81.05% on CoNLL-2003, outperforming SpaCy (54.33%), Gate (53.55%), OpenNLP (42.18%), and NLTK (48.47%) under identical experimental conditions. Thus, if NER accuracy is the top priority, StanfordNLP is the best choice. However, Schmitt et al. [98] also point out that the advantage of StanfordNLP narrows from roughly 30% on CoNLL-2003 to 15% on the GMB corpus, a domain on which it was not trained, suggesting that SpaCy is better suited for texts from domains other than news, as the performance gap with StanfordNLP narrows significantly on out-of-domain corpora [98].

Transformer-based models achieve much better performance for domain-specific KG construction. Yang et al. [126] report that BERT-BiLSTM-IDCNN-CRF achieves 89% precision, 87% recall, and 88% F1 on geological NER, outperforming Bert-BiLSTM-CRF (77% F1) and HMM (67% F1) on the same dataset. This pattern has been observed in specialized domains: transformer models trained on domain-specific data always outperform general tools and are preferable for healthcare KG construction when domain-specific training data is available [10, 125].

LogMap [102] achieves F-scores of 0.825-0.870 on large biomedical ontologies and completed matching of Mouse Anatomy and NCI Anatomy in 15.7 s when compared to 10-23 min for competing tools in the OAEI evaluation making it preferable for large-scale biomedical ontology alignment tasks. COMA++ [103] offers complementary strengths through its composite matching strategy, making it suitable for general heterogeneous schema alignment across diverse non-biomedical sources where domain-specific reasoning is not required.

5.4. Knowledge fusion

This step aims at merging diversified knowledge sources to form a complete and consistent graph by resolving duplicates and conflicts, completing missing links, normalizing the values, and providing knowledge inference. This involves schema/ontology alignment, entity alignment, triple-level conflict detection and resolution, and rule-based fusion, which are integrated with embedding-based methods to enhance scalability. The traditional approaches used lexical and structural similarity measures, whereas Entity Alignment methods based on representation learning have better performance and efficiency [110]. In domain-specific knowledge graphs, word embeddings are used to predict the RDF triples for entity completion problems. Knowledge fusion is widely used in intelligent data processing and retrieving hidden knowledge patterns from knowledge graphs.

5.5. Graph construction

Graph building aims at transforming the structured data schema to a graph database schema, which includes entities, relationships, attributes, and properties. Structured data usually has a table for every entity and relationship wherein each entity table becomes a node in the graph schema, every row in the entity table forms a one-node instance having a global unique identifier and has a set of properties referring to the column values, and every row in the relationship table becomes an edge connecting two entity nodes. Graph indexing mechanisms are used to organize the nodes, relationships, and properties to enhance the query performance, especially when dealing with large-scale data in graph databases.

5.6. Knowledge retrieval, validation, and reasoning

Once the KG has been deployed, knowledge retrieval mechanisms such as SPARQL [6] or Cypher [7] queries can be used to retrieve efficient access from the structured information. Knowledge validation ensures the correctness or the factuality of the generated answer, and periodic updates maintain the graph's consistency over time. Knowledge reasoning in a knowledge graph is to obtain new facts based on existing ones by reducing negative events based on familiar patterns. This step helps to check the consistency of a knowledge graph, such that it does not include logically impossible statements, and to complete the

knowledge graph if any missing links exist. Traditional approaches like ontology or rule-based symbolic reasoning can be complex on very large knowledge graphs. The recent advancements in Graph Neural Networks (GNN)-based reasoning and Neuro-symbolic reasoning [111] better capture the graph structure. SPARQL reasoning helps to identify new facts based on the rules using Resource Description Framework (RDF)/ Web Ontology Language (OWL). LLMs exhibit better performance in KG construction and reasoning tasks like link prediction, path reasoning, and rule extraction when given in a structured context [112].

6. Multimodal knowledge graphs

Traditional knowledge graphs organize textual data in a structured representation, whereas a multimodal knowledge graph integrates semantic and heterogeneous information, which includes image, audio, video, sensor data, and gives them a graph structure. Multimodal knowledge graph (MMKG) integrates multiple modalities and numerical attributes with structured relational data, enabling enhanced link prediction and cross-graph entity alignment [113].

MMKG Construction can be done with two architectures: Firstly, Entity-centric MMKG, where each entity node contains textual descriptions and audio and video embeddings, and the multimodal features, given as properties of the node. Secondly, modality-specific subgraphs, here in each modality, for example, the Image modality will have an Image graph, and the text modality will have a text graph and audio and video graphs, respectively, for audio and video modalities. Entity alignment techniques are used to connect the corresponding entities of different modalities with edges. Fig. 10 illustrates the

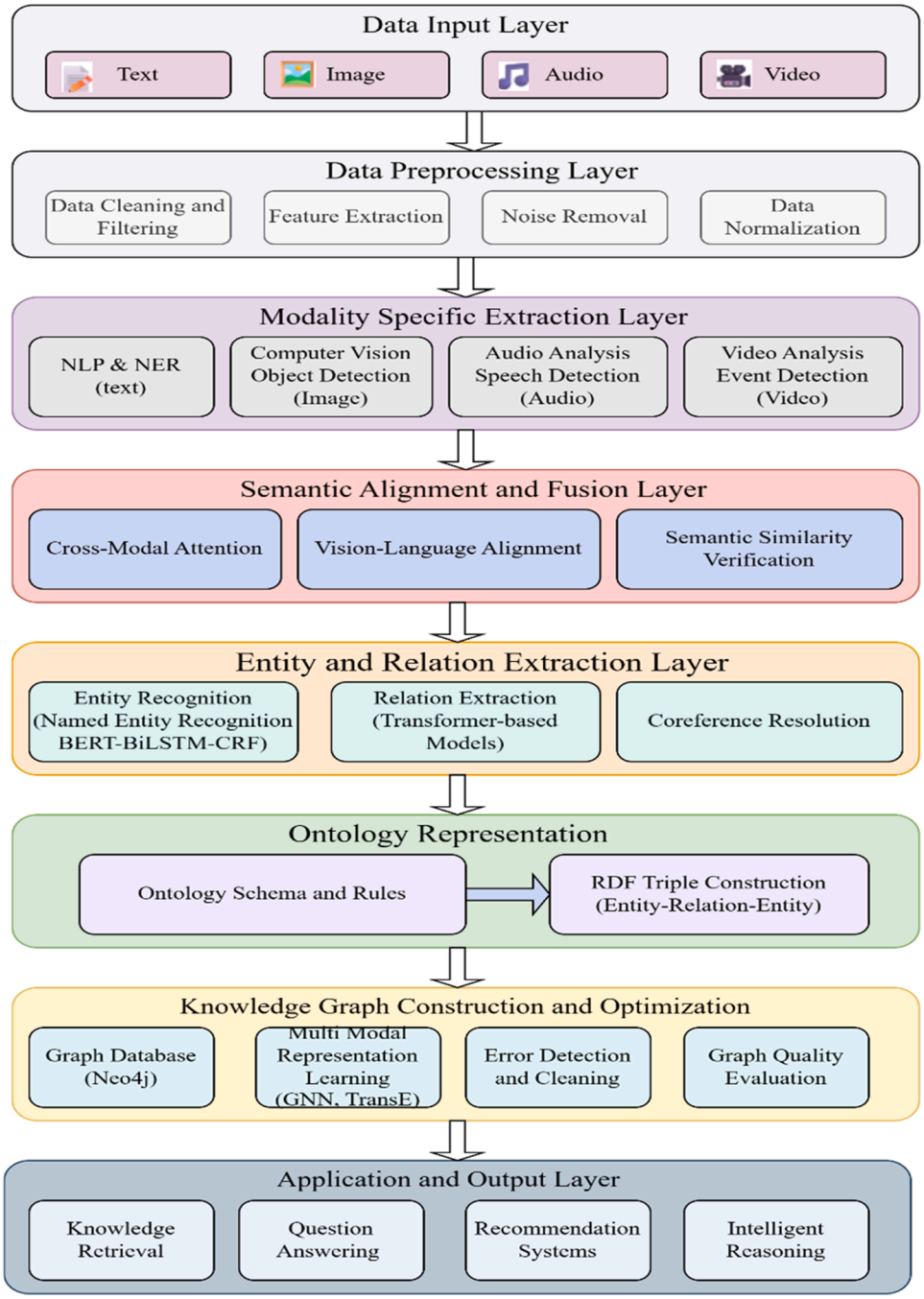


Fig. 10. Multimodal Knowledge Graph construction pipeline.

Multimodal knowledge graph construction pipeline, showing the flow from raw multimodal data through entity alignment, semantic fusion, and graph instantiation. This multi-stage pipeline is discussed below:

6.1. Data collection and preprocessing

This step starts with collecting the structured KGs from General-purpose knowledge graphs like Wikidata, DBpedia, YAGO, Freebase, etc., and from domain-specific structured KGs like Unified Medical Language System (UMLS), Systematized Nomenclature of Medicine – Clinical Terms (SNOMED-CT), and from Neural/symbolic extraction pipelines. The collected KGs are then preprocessed to remove inconsistencies and duplicate data, normalize the data, and check for any missing links that can lead to an incomplete graph. This can be done using Python tools and libraries.

6.2. Entity alignment and entity linking

As there are different modalities of data which consists of entities that may be linked to each other, a cross-modal entity linking is crucial and is done at this stage. Modality-specific extraction layer extracts semantic units from each modality using techniques such as event detection for video, object detection for images, speech recognition for audio, and NLP-based NER for text. For example, it is difficult to decide that an entity named Jaguar should be linked to a Car or an Animal. But, with the help of an image related to a wild animal, it can be concluded that here, the entity Jaguar is to be linked with Animal. We use vision-language models like Contrastive Language–Image Pretraining (CLIP) [114] to match the entities. Multimodal Entity Linking (MEL) datasets have been very useful for multimodal entity linking tasks. After much research has been conducted for a good MEL dataset, a WIKIDiverse dataset, and a manually annotated dataset having diversified topics and entity types was presented [115].

6.3. Semantic alignment and fusion

In MMKG, different modalities generate different embedding representations. To map heterogeneous modality-specific embeddings into a common semantic representation space, this step plays a crucial role while constructing MMKG. Semantic correlations between text, image, audio, and video modalities are captured using cross-modal attention mechanisms and vision-language alignment models. To have entity consistency across modalities, semantic similarity verification methods such as cosine similarity and embedding-based matchings are used. The aligned representations are then fused using attention-based or GNN (Graph Neural Network)-based fusion representations to create integrated multimodal entity embeddings for tasks like entity linking, relation extraction, and graph completion.

6.4. Feature extraction

Once the modality-specific embeddings are aligned, tasks like entity recognition, relation extraction, and coreference resolution take place. Entity recognition is done using NER to detect entities in text. As the entities are aligned, now the system can confirm whether an image corresponds to a textual entity. Relation extraction is done at this step using transformer relation extraction models to capture cross-modal relations. If multiple mentions are referring to the same entity, coreference resolutions help to refer to a particular entity with a pronoun at a later stage. For example, “Dr. Ethan” and “he” refer to the same entity based on the context they are used in. Table 8 refers to transformer-based techniques that can be used to convert the raw data to a machine-understandable format and can be used for entity linking, knowledge fusion, KG completion, and reasoning.

Table 8
Transformer-Based Models to Convert Raw Data to Machine-Understandable Format.

Type	Techniques
Text Data	BERT [10], RoBERTa [12], LLM Encoders
Image Data	Vision Transformer [116], CLIP Visual Encoder [117]
Audio Data	wav2vec [118], MFCC extraction
Video Data	I3D [119], Video Transformers, CLIP-based video encoders [114]
Time-Series Data (Sensor Data)	TS2Vec (Time Series to Vector representation model) [120], LSTMs [121], Autoencoders

6.5. Graph construction

The normalized preprocessed KGs with multimodal embeddings are now being converted to actual KG nodes(entities), edges(relationships), and properties by following an ontology schema. For ontology schema creation, we use tools like OWL/ Protégé and Neo4j, which give a clear idea about what type of nodes and relationships exist and what properties entities have. For example, a person XYZ is a text-data, heart rate captured by a Fitbit device is a time-series data, an ImageNode depicts the Image modality, and an ECG Signal node derives the sensor data. Once these are finalised, we can create entities and relationships using Neo4j, Graph ETL(Extract-Transform-Load) frameworks, and each entity has properties that are extracted from the knowledge base (structured data).

6.6. KG construction

Using multimodal embeddings, entities and relations extracted, and ontology being mapped, the triples can now be converted into a structured multimodal KG. As it is a graph, entities are converted to nodes, relations are converted to edges, and modality-specific embeddings are stored as feature vectors of associated nodes. The graph is then deployed to Neo4j, a graph database system, which assigns unique identifiers to every node, and constraints are applied on relations to enable efficient query processing. Indexing mechanisms are applied, and nodes are associated with their multimodal embeddings, transforming structured knowledge from a knowledge base to a queryable graph structure.

Translational embedding models such as TransE and ComplEx understand the low-dimensional vector representation of entities and relations and enable link prediction, entity matching, and KG completion. To improve link prediction and generalization in multimodal contexts, a novel inductive MMKG model [122] is presented that modifies embeddings using class-adaptive properties. GNN models are used to understand the higher-order structural dependencies in the graph and learn context-aware node representations, improving reasoning and link prediction accuracy.

The next step is to ensure correctness and factuality by enabling error detection and cleaning the graph. At this step, deduplication of entities is done, conflicting triples are identified, and constraint validation is enforced to verify that the relationships connect to the correct entities to improve reliability and semantic correctness. Next, to evaluate the effectiveness of the constructed MMKG, evaluation metrics are applied. The detailed overview of these metrics is provided in Section 10 of this paper.

6.7. KG completion

Using multimodal embeddings and the knowledge graph created, we can predict the missing links, fill incomplete attributes, and structure the cross-modal data using multimodal ML models, like TransE, ComplEx and MMKGC-Transformer. Recent research presents a unified KG representation learning framework [123] that improves embedding quality and link inference capabilities in complex knowledge graphs, using flexible algebraic structures to model entities and relations. Once the KG is validated and completed, it is used for knowledge retrieval and can be

used in various applications like Question-Answering systems, Recommendation systems, and in Intelligent reasoning tasks.

6.8. Illustrative example of MMKG construction

We take a biomedical MMKG construction scenario as an example of cross-modal entity linking. A clinical report says: "Patient has consolidation in the right lower lobe consistent with pneumonia." BioBERT [10] extracts textual entities: {Patient, Consolidation, Right Lower Lobe, Pneumonia}. The corresponding chest X-ray image produces visual entities via ViT [116]: {OpacityRegion, LungRegion}. Then, CLIP [117] calculates cross-modal similarity to establish the correspondence: (OpacityRegion [image], sameAs, Consolidation [text]). The generated cross-modal triples are: (Patient, hasCondition, Pneumonia) (Pneumonia, hasVisualEvidence, OpacityRegion) (OpacityRegion, locatedIn, RightLowerLobe) This example demonstrates the joint representation of textual and visual modalities as structured triples in an MMKG.

6.9. Cross-modal alignment challenges

Despite great advances in the construction of multimodal KGs, the integration of heterogeneous modalities presents unique challenges that do not exist in unimodal KG pipelines. The construction of MMKG is affected by three fundamental challenges: modality gap, missing modalities, and noisy alignment, which need to be explicitly addressed in the construction process [113].

6.9.1. Modality gap

One of the key challenges in building MMKGs is the modality gap, which is the representational gap between the embeddings from different modalities. Liang et al. [149] demonstrate that even when vision-language models such as CLIP [117] are trained on paired image-text data, their embedding spaces are still geometrically separated, meaning that text embeddings and image embeddings occupy separate regions of the representation space, rather than in a unified joint space. In MMKG construction, this results in systematic misalignment of visual entity representations with their textual counterparts. An image of a medical scan and its corresponding clinical description may be encoded in geometrically distant regions of the embedding space despite referring to the same entity. To bridge this gap, specialized alignment architectures are required, such as cross-modal attention mechanisms and contrastive learning objectives that explicitly minimize representational distance between paired modalities [117].

6.9.2. Missing modalities

In the real world, the construction of MMKG often encounters the scenario of missing modality, where some entities have complete multimodal information while others are missing one or more modalities. In Richpedia [90], images are not available for all entities, and in TIVA-KG [91], audio and video representations are available only for a subset of the entities. Missing modalities give rise to structural inconsistency in the KG — entities with partial modalities cannot be directly compared with fully represented entities during embedding or reasoning tasks. Recent work tackles this by prototype-based inductive learning that adaptively embeds MMKG entities regardless of the completeness of modalities [122]. This remains an open challenge for large-scale MMKGs where missing modalities are ubiquitous.

6.9.3. Noisy alignment

Noisy alignment refers to the existence of incorrect correspondences among automatically matched cross-modal entity pairs, which is a persistent problem in large-scale MMKG construction where manual verification of millions of image-entity links is infeasible. IMGpedia [64] builds image-entity links by automatically matching Wikimedia images to Wikipedia entities, which inevitably introduces noise when visually similar but semantically different entities are wrongly linked. WIKIDiverse [115] further points out the challenge of diverse contextual topics in multimodal entity linking, where one image could plausibly be linked to multiple entities. Noisy alignment propagates errors across the KG—one incorrect image-entity link can corrupt downstream cross-modal reasoning tasks [113]. To mitigate noisy alignment, confidence-based filtering and robust embedding methods that are resilient to alignment noise are required.

7. Construction of domain-specific knowledge graph based on healthcare data

The healthcare industry generates massive volumes of structured and unstructured data through Electronic Health Records (EHR), Electronic Medical Records (EMR), lab reports, prescriptions, clinical notes, medical imaging, and public medical datasets. However, these datasets are often fragmented across multiple sources and formats. Hence, extracting meaningful insights becomes difficult using traditional relational database systems alone.

A Domain-Specific Knowledge Graph acts as a powerful solution to structure healthcare data in the form of entities (such as patients, diseases, medicines, lab tests, doctors) and relationships (such as diagnosedWith, prescribed, hasLabTest, consultedBy). The KG enables advanced querying, reasoning, clinical decision support, and semantic interoperability. The architecture of a domain-specific KG consists of a multi-stage pipeline. Fig. 11 depicts that each stage performs a critical task in transforming raw healthcare data into a structured, queryable, and meaningful knowledge graph.

The architecture is divided into seven major layers:

7.1. Healthcare data sources (Input)

Healthcare domain data can be collected from various sources: Structured data sources include hospital databases (patient records), lab datasets (test values), prescription datasets, Insurance datasets, and disease registries. Semi-structured sources include JSON/XML health records and API-based datasets. Unstructured sources include clinical notes, discharge summaries, doctor reports, medical articles & web pages. In general domain KG construction, data is freely available, but healthcare data sources are subject to strict access controls and institutional review board (IRB) approval requirements since patient records contain sensitive protected health information (PHI) regulated by HIPAA and GDPR regulations [94].

7.2. Data collection and standardization

In this stage, data is gathered from datasets and standardized into a consistent format. Key tasks here include identifying important healthcare tables such as (Patients, Visits, Diagnoses, Prescriptions, Lab Tests, Lab Results), standardizing column names (example: patient_id, visit_id), and converting date formats into YYYY-MM-DD, ensuring consistent naming (example: Type 2 Diabetes instead of diabetes mellitus). Entities like diseases and drugs are extracted using biomedical NER models like BioBERT, ClinicalBERT, and MetaMap. This stage ensures that the data is ready for processing without inconsistencies.

Crucially, clinical data collection in healthcare differs from general domain KG construction in that raw EHR data cannot be directly processed, and a mandatory de-identification step must be performed before any downstream processing to remove patient identifiers such as names, dates, and geographic information before any NLP pipeline can be applied [94,129]. This de-identification layer is totally absent from general domain KG construction pipelines and adds a source of error when identifiers are wrongly removed or retained.

7.3. Data preprocessing

Healthcare datasets often contain missing values, duplicates, and

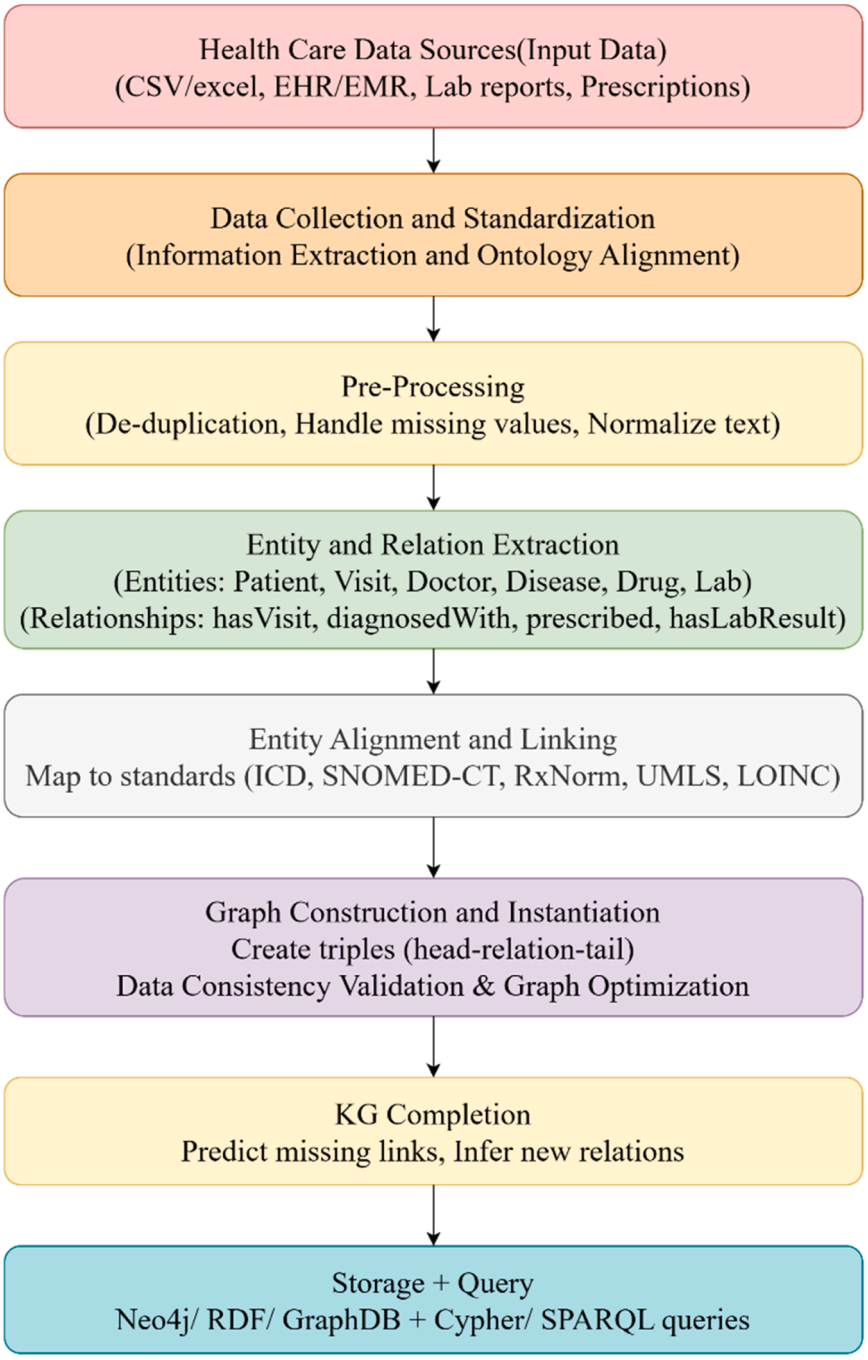


Fig. 11. Domain-Specific Healthcare Knowledge Graph construction pipeline.

noisy information. Hence, pre-processing is essential. Major pre-processing activities include handling missing values, where missing values are usually replaced with numerical values with mean/median if required, and null rows are also removed if incomplete or irrelevant, deduplication of data, where patient duplicate records and duplicate visit entries are removed to eliminate duplication of data. Text normalization is an important step, as here we tend to standardize diseases and medicine names with ICD (International Classification of Diseases) codes [124]. Data type validation is crucial, where the type of data should be validated, such as age should be numeric, and lab results should be numeric (where applicable). This step improves data quality and reduces KG errors.

However, healthcare preprocessing faces a unique challenge not present in general KG pipelines — temporal inconsistency across EHR systems, where the same patient condition may be recorded with different timestamps, terminology, and coding conventions across multiple hospital visits [129,130]. This temporal heterogeneity necessitates dedicated preprocessing to temporally align sequences of clinical events, such that reliable results of entity extraction can be obtained.

7.4. Ontology schema design

Domain-specific KGs require an ontology representation to design the structure of the healthcare KG. The ontology contains the core entities, here specified as E = {Patient, Visit, Condition, Medication, LabTest, LabResult, Doctor, Hospital}, along with their relationships, here specified as R = {hasVisit, consultedBy, visitedAt, diagnosedWith, prescribed, hasLabResult, testType}. Every relation is specified with

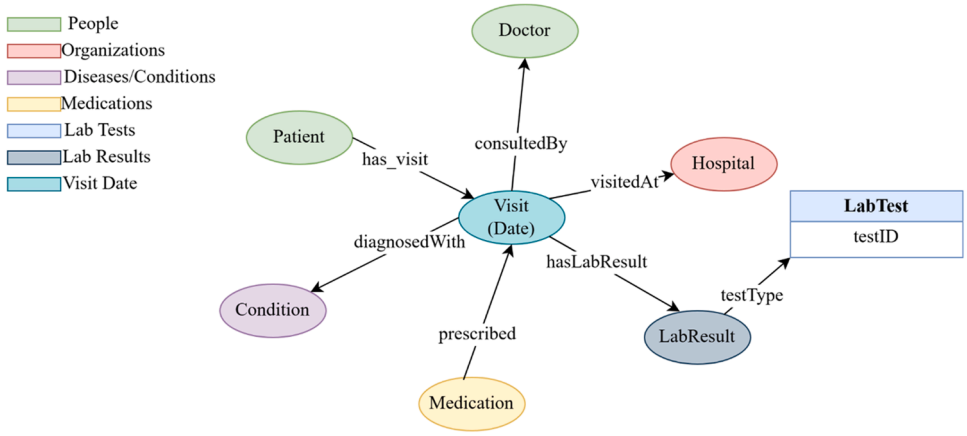


Fig. 12. Ontology Schema of a Healthcare Knowledge Graph.

domain and range constraints to enable consistency and validity in the graph. As shown in Fig. 12, the visit node that acts as the central node has date associated with it, connecting the patients to real-world clinical workflows in a structured manner. The ontology design acts as a formal foundation for entity extraction and entity alignment in the graph construction process.

Designing healthcare ontology is much more restricted than general domain ontology design, as it must satisfy multiple established medical standards at the same time, for example, ICD-10 [124] for disease classification, SNOMED-CT [50] for clinical terminology, UMLS [51] for cross-terminology mapping, etc. The task of mapping these standards to a common ontology schema, preserving semantic consistency, is a methodological challenge specific to the healthcare domain that general ontology design for KG does not face.

7.5. Entity and relation extraction

Entity extraction identifies the core healthcare elements, while relation extraction defines meaningful relationships between those entities. Named Entity Recognition (NER) identifies domain-specific entities from text. Entities extracted for the healthcare domain are shown in detail with their properties in Table 9, stated below.

Relationships define how entities are connected. Table 10 depicts the core relationships in healthcare KG and is presented in triple representation following RDF standards. Entity and relation extraction marks the foundation for graph building. Modern approaches use BERT-based or transformer models for high-accuracy entity recognition [125]. The BERT-BiLSTM-CRF model has demonstrated F1 scores above 88% for medical entity extraction [126]. In entity relation extraction, transformer-based deep learning models such as RoBERTa have achieved accuracy rates of 85-90% with proper training and multi-task learning [127]. Medical entity extraction is more challenging than general domain NER as clinical abbreviations are highly ambiguous (e.g., "MS" can refer to multiple sclerosis, mitral stenosis, or morphine sulfate depending on clinical context), and a single condition can be described by dozens of synonymous terms in different physicians' notes [26,125]. Moreover, an entire institutional corpus may have fewer than ten documents for rare disease entities, leading to severe data sparsity that general domain extraction models are not designed to handle [52].

Table 9
Healthcare KG Entities and Key Properties.

Entity	Key Properties
Patient	patient_id, name, age, gender, city, blood_group
Visit	visit_id, visit_date, hospital, doctor
Condition (Disease)	condition_id, condition_name, ICD-10 code, severity
Medication (Drug)	drug_id, drug_name, drug_class
Prescription	p_id, dose, frequency, duration_days
LabTest	test_id, test_name, unit
LabResult	result_id, value, test_id, visit_id
Doctor	doctor_name, specialization (optional)
Hospital	hospital_name, location (optional)

Table 10
Healthcare KG Relationships.

Head Entity (Subject)	Relation (Predicate)	Tail Entity (Object)
Patient	hasVisit	Visit
Visit	consultedBy	Doctor
Visit	visitedAt	Hospital
Visit	diagnosedWith	Condition
Visit	Prescribed	Medication
Visit	hasLabResult	LabResult
LabResult	testType	LabTest

7.6. Entity linking and alignment

Entity linking maps the extracted entities to standardized medical identifiers in restricted vocabularies. Entities are extracted and mapped to standardized vocabularies such as ICD-10 [124], SNOMED-CT [50], and UMLS [51]. Standard medical vocabulary linking ensures that the clinical entities are standardized and mapped for the diagnosis purpose. For example, if a raw text is given like "Patient is diagnosed with sugar disease", the entity extracted will be "sugar disease" and this will be linked to standardized medical terminology with unique identifiers, like "Diabetes Mellitus→ICD-10: E11" and "SNOMED CT: 44054006". In the healthcare domain, entity linking is required to link extracted clinical mentions to standardized medical codes through a process known as medical concept normalization. This process involves resolving abbreviations, synonyms, and misspellings to controlled vocabularies such as UMLS [51] and SNOMED-CT [50]. This is very different from general domain entity linking, where disambiguation through Wikipedia or Wikidata is enough [110].

Entity alignment ensures consistency across multiple datasets and removes duplication of data. It ensures that all representations of the same concept map to the same node in a KG. It resolves cross-database matching and aligns all matching entities to the same node. For example, the clinical terms, "High BP" and "Hypertension", will be standardized with an ICD-10: I10. In the same way, the terms "DM" and "Diabetes Mellitus" will be standardized with an ICD-10: E11. This stage improves graph correctness and prevents duplication of nodes.

7.7. Graph construction and instantiation

In this stage, the structured data is converted into a graph format. As the entities are mapped to standards using entity linking, here, RDF-style triples are created as head entity, relationship, tail entity. Once the

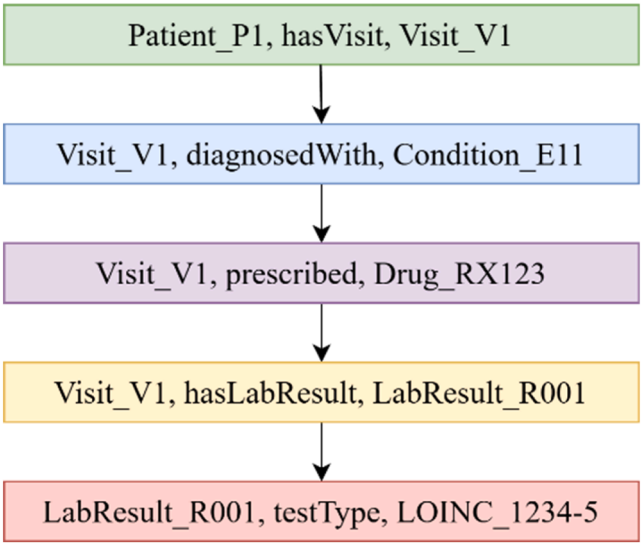


Fig. 13. RDF Triple Construction in Healthcare KG.

triples are constructed, graph database instantiation is done, where these triples are loaded into Neo4j, a property graph model with Nodes as Patient, Visit, Condition, Medication, LabTest, LabResult, Doctor, Hospital, and Edges as hasVisit, consultedBy, visitedAt, diagnosedWith, prescribed, hasLabResult, testType. Once the nodes and edges are loaded, unique constraints are created, and domain-range restrictions are imposed. This is done to ensure correctness for the created healthcare KG. After the graph is created, data validation is done to remove duplicate nodes and check for any incorrect ICD codes or missing relationships to ensure graph reliability. In the latter part, graph optimization is done using graph embeddings like TransE, ComplEx, and GNNs to improve link prediction accuracy. A visual KG, "disease-prescription-medication" KG using KG technology was built in the Neo4j graph database, providing diagnostic decision support and treatment recommendations for clinical practice [128]. An electronic health record KG is built using multicenter fragmented patient medical data for collaborative reasoning while safeguarding data privacy through semantic reasoning and a blockchain-enabled network [129]. Data-driven medical reasoning is showcased in one of the foundational works where a large-scale healthcare KG was constructed from electronic medical records of real-world hospital data to discover disease co-occurrence and clinically meaningful patterns [130]. Fig. 13 showcases the complete flow of a patient visit to diagnosis, medication prescribed, and its lab observations that are modeled by a knowledge graph.

Healthcare KG construction additionally needs to encode the temporal ordering of clinical events – a requirement that is missing in general KG construction. Patient visits, diagnoses, prescriptions, and lab results are represented as time-stamped nodes and edges that preserve the temporal sequence of the patient journey [129,130]. This temporal graph structure bears resemblance to the time-series KG principles in event KGs such as GDELT [66] and ICEWS [67], but is adapted to the clinical domain where temporal ordering directly affects diagnostic and treatment inference.

7.8. KG completion

KG completion predicts missing links in the KG by learning statistical patterns from training data. For example, for a diabetes patient, based on learned relational patterns in the graph, a doctor prescribes Metformin, and if one diabetes case has a missing prescription, KG completion may predict a triple, such as (Visit, prescribed, Metformin). Another example that shows how KGs infer new triples can be stated as, if the KG has (Patient_P1, diagnosedWith, Diabetes) and (Diabetes, treatedWith, Metformin), a completion model may predict (Patient_P1, prescribed, Metformin). Tools of KG Completion include Embedding-based models (TransE, ComplEx), Graph neural networks, and similarity-based scoring techniques that rank entities based on learned representations in data.

KG completion for healthcare is riskier than general domains since a hallucinated drug-disease relation or an incorrectly inferred diagnostic relation may directly influence clinical decision support systems [112, 130]. This requires clinical expert validation of LLM-generated completions before adding them to production healthcare KGs, a validation step not required for general domain KG completion.

7.9. Storage and query layer

After construction, the KG is deployed in a graph database, such as Neo4j (Property Graph), RDF Graph stores, GraphDB / Blazegraph, and Queries can be performed using Cypher [7] (Neo4j) and SPARQL [6] (RDF graphs). This enables consistent storage, indexing, and querying of structured clinical data for real-time knowledge retrieval. Here, in Fig. 14, a sample healthcare KG construction is shown. The construction of a domain-specific healthcare KG involves converting healthcare data into a structured graph format that represents real-world relationships among patients, diseases, medications, lab tests, and visits.

Storage systems for healthcare KG and also implement role-based access control, audit logging to meet HIPAA compliance [94], allowing access to patient-level graph data only for authorized clinical personnel. This regulatory aspect of storage architecture is entirely missing from the generic domain considerations of the deployment of KG.

7.10. Key challenges and methodological differences in healthcare KG construction

Whereas Sections 7.1–7.9 discuss the pipeline for the healthcare KG, every step has specific challenges that make the process different from other domain pipelines. The current section identifies four basic challenges that are unique to the healthcare KG pipeline, and Table 11 compares methodological differences between general and healthcare-specific KG construction pipelines.

7.10.1. Challenge 1–Privacy and de-identification

Privacy and regulations are the most basic constraints that distinguish the creation of healthcare KGs from general domain pipelines. Protected information in the USA is covered by the HIPAA regulation, and GDPR in Europe, which require the de-identification of protected health information to be processed at all [94]. In contrast to the general domain KG pipelines, where texts can be accessed freely, healthcare KGs require the de-identification module to be integrated as the initial stage of processing. Any errors introduced during de-identification may propagate to the resulting KG in the form of incorrectly identified entities [129].

7.10.2. Challenge 2–Temporal reasoning in EHRs

Electronic health records are inherently temporal – patient conditions, medications, laboratory results, and diagnoses are in a constant state of flux over time. Static KG triples are not sufficient to model clinical event sequences where the temporal ordering plays a direct role in diagnostic inference [129,130]. This requires incorporating time-series KG principles that are used in event KGs such as GDELT [66] and ICEWS [67] into the clinical domain. Standard subject-predicate-object triples are enriched with timestamps and causal temporal relations, which are not required by generic KG construction pipelines.

7.10.3. Challenge 3–Multi-source heterogeneity

Healthcare data is generated by a variety of heterogeneous sources—structured EHR databases, unstructured clinical notes, medical imaging reports, laboratory systems, and administrative claims, each with its own terminologies and coding systems [94]. The integration of

Table 11
Methodological differences between general and healthcare-specific knowledge graph construction.

Aspect	General KG Construction	Healthcare KG Construction
Data access	Publicly available text	Restricted EHR data requiring IRB approval [94]
Pre-processing	Standard NLP pipeline	Mandatory de-identification before extraction [94,129]
Ontology	Single domain ontology	Multi-ontology alignment — ICD-10, SNOMED-CT, UMLS [50,51,124]
Temporal modeling	Secondary consideration	Primary requirement for clinical event sequences [129,130]
Entity ambiguity	Named entity disambiguation	Medical concept normalization to standard codes [51]
Validation	Automated benchmark evaluation	Clinical expert validation required [130]
Data sparsity	Manageable for popular entities	Severe for rare diseases and conditions [52]
Regulatory constraints	None typically	HIPAA and GDPR compliance mandatory [94]

these sources requires a simultaneous ontology alignment across ICD-10 [124], SNOMED-CT [50], and UMLS [51], a methodological complexity well beyond general KG construction, where a single ontology generally governs knowledge representation.

7.10.4. Challenge 4–Data sparsity in rare diseases

In healthcare KGs for rare diseases, data sparsity is a major problem because the patient populations are small and clinical documentation is scarce [52]. LLM-based KG completion methods [112] that perform well on commonly documented conditions may hallucinate clinical relationships for rare diseases where the training signal is limited – a risk with direct patient safety implications that makes validation by clinical experts mandatory before any LLM-generated completions are incorporated into production healthcare KGs [130].

Together, these four challenges explain why healthcare KG construction cannot simply take pipelines from general domains without significant domain-specific adaptation and why the quality of each pipeline stage directly determines the KG's reliability in the downstream clinical decision support systems.

A simplified representation of the healthcare KG is shown below:

This enables linking medical history, helps identify often co-occurring conditions, analyzing prescribed medication and treatment patterns.

8. Integration approaches for KGs and LLMs

To create AI systems that are factually correct and confidently strong, there is a need to integrate KGs and LLMs. LLMs strong capability of learning through statistical patterns from a large text corpus makes it best for delivering strong response generation, summarization and reasoning tasks. Nevertheless, the lack of built-in fact-checking of the responses generated, which led to hallucinations, is the major drawback they suffer from. While KGs offer high precision, interpretability, and provide strong reasoning capabilities that enhance the performance of LLMs. By integrating these two paradigms, AI systems can be more dependable and can offer explicable intelligence.

Recent research broadly divides KG+LLM integration into three paradigms, namely: KG-enhanced LLMs, LLM augmented KG and Synergized LLMs+KGs [131]. While Pan et al. [131] proposed the base three-paradigm taxonomy for KG-LLM integration, three limitations motivate the extensions proposed in this survey: First, their taxonomy does not provide guidance on which KG type is most suitable for which integration paradigm. Second, it does not address how KG construction quality affects integration effectiveness. Third, it predates GraphRAG [137] and HybridRAG [134], which represent practically significant retrieval implementations beyond their three paradigms. Fig. 1 and Table 12 present the analytical framework that addresses all three limitations. A deeper understanding of the categorization shown in Fig. 15 is discussed below.

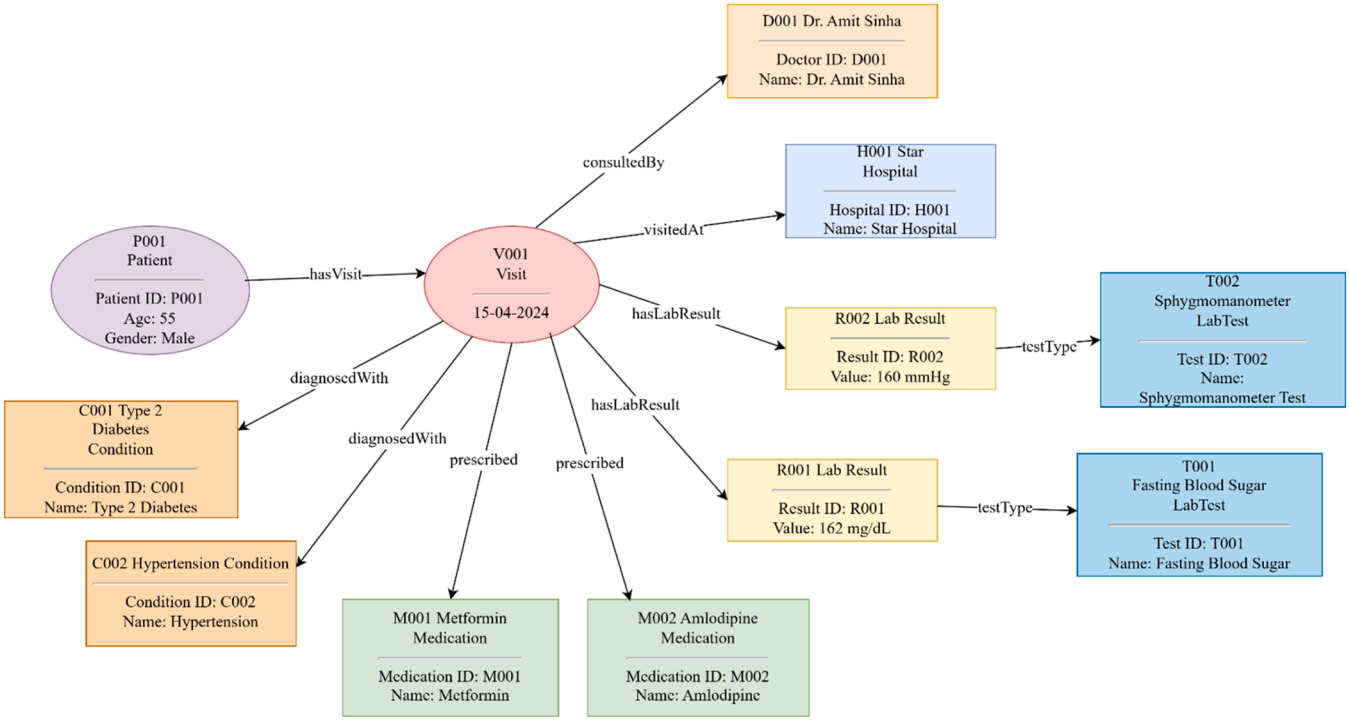


Fig. 14. Sample Healthcare KG connecting Patients, Diseases, Medications and Clinical visits.

Table 12
Comparative analysis of LLM-KG integration strategies.

Strategy	Core Mechanism	Applicable Condition	Key Limitation	Referenced Work
KG-enhanced LLM (Static)	KG facts embedded into LLM parameters during pre-training via entity-aware masking	Domain knowledge is fixed and well-established; model retraining is feasible	Cannot update without full retraining, prohibitively expensive for large KGs	ERNIE [133], KEPLER [136]
KG-enhanced LLM (Dynamic)	KG subgraphs retrieved at inference time via graph traversal or embedding search	Open-domain tasks where knowledge evolves over time and retraining is not feasible	Retrieval quality bounded by KG completeness and freshness	GNN-RAG [135]
LLM-augmented KG	LLM generates missing triples, completes entities, and constructs ontologies from text	Incomplete KG with rich associated textual sources available	Hallucinated triples degrade KG quality, error propagation into downstream tasks	Zhu et al. [112], Ibrahim et al. [132]
Synergized LLMs+KGs	Bidirectional enhancement — KG structures LLM reasoning while LLM updates and extends KG	Complex multi-hop reasoning requiring both neural and symbolic inference	High computational cost; complex pipeline design and maintenance	Pan et al. [131], DeLong et al. [111]
GraphRAG	Community detection on document-derived KG with hierarchical summarization for global retrieval	Global thematic query understanding across large heterogeneous document collections	Performance is bounded by the quality of the automatically constructed graph index	Edge et al. [137], Peng et al. [108]
HybridRAG	Fusion of dense vector retrieval with structured KG graph traversal in a unified pipeline	Enterprise settings with mixed structured relational and unstructured text queries	Retrieval pipeline complexity increases latency; requires careful tuning	Sarmah et al. [134], Peng et al. [108]

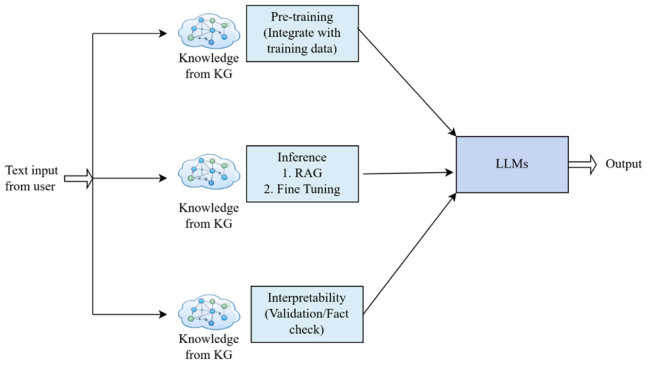


Fig. 15. KG–enhanced LLM paradigm.

8.1. KG – enhanced LLMs

To enhance the performance of LLMs we here integrate KG into LLMs, which can solve issues like hallucinations and lack of reasoning. To achieve this, entities and relations from a KG can be represented as continuous space vectors that an LLM can use directly during training or inference [132]. Here, the KG supports the model during pre-training, where LLM learns the knowledge of a KG or during inference by serving as an external, structured knowledge source, from which LLM retrieves the information to perform response generation for domain-specific queries. KG-enhanced LLMs can be categorized into three phases:

8.1.1. KG integration during pre-training

Incorporating KGs into LLMs during the pre-training phase. Here, an LLM is trained on more knowledge entities, knowledge-guided masking, and unified learning of text and KG triples rather than just raw text. This captures deterministic factual knowledge and explicit entity relationships. KGs can be integrated into LLM inputs by introducing similar knowledge sub-graphs into the inputs of LLMs. A triple is broken down as a token sequence and added to sentences by ERNIE 3.0 [133]. In order to integrate knowledge with textual representations, it either hides the relation token in a triple or the tokens in the sentences where the entities were added. This technique helps the model to encode explicit entity relationships and helps to improve the factual recall. The other technique is KGs Instruction-tuning, where instruction-tuning datasets are created using facts and the structure of KGs. Here, LLMs are fine-tuned to better understand the structure of KGs and the user's instructions to perform complex tasks. Fine-tuning involves extracting entities and relations using Named Entity Recognition (NER) and Relation Extraction techniques. Wherein the extracted entities and relations are embedded into continuous space vectors using methods like node2vec and Graph Neural Networks (GNNs), which allows LLMs to embed structured knowledge during training or inference.

8.1.2. KG – aware inference

This is a more practical way of integrating KGs at inference time. The term "inference" means to generate outputs or predictions on new input data based on pre-trained data. The real-world data keeps changing and evolving every day, which requires constant real-time knowledge updates. This is implemented using Retrieval Augmented Generation (RAG), where non-parametric and parametric modules are combined to handle external knowledge bases. RAG retrieves the relevant knowledge from its vector databases and feeds it into LLMs. RAG pipelines have shown exceptional results in reducing hallucinations and improving factual consistency. Multi-hop reasoning can also be enhanced by KG-guided reasoning methods, which require the model to conduct n number of hops by traversing entity relations as intermediate reasoning steps to reach a factual answer.

A new retrieval-augmented generation framework called HybridRAG [134] gives large language models a more relation-aware and semantically coherent context by fusing structured knowledge graph traversal with dense vector retrieval. Compared to conventional RAG techniques, HybridRAG enhances factual grounding and multi-hop reasoning capabilities by combining the advantages of both neural similarity search and symbolic graph exploration.

GraphRAG extends classical RAG by constructing a graph-based index in two steps: first, by generating an entity knowledge graph from source documents, and then by pre-creating community summaries of strongly related entity groups. At query time, partial responses are generated from each relevant community summary and aggregated into a final response. In a GraphRAG pipeline, entity linking is used to find relevant KG nodes, and then a multi-hop graph traversal is performed. The retrieved subgraph is then converted into graph-aware embeddings or serialized into structured prompts to the LLM to enable relational reasoning via traceable reasoning paths.

A GNN-RAG framework that uses lightweight Graph Neural Networks (GNNs) for effective and efficient graph retrieval is introduced in Graph enhanced reasoning [135]. This framework improves retrieval performance by effectively handling context from in-depth parts of the graph. Similar approaches like KEPLER [136] produce text-enhanced knowledge embeddings to enable structured knowledge influence token embeddings during inference by integrating factual knowledge into pre-trained language representation models. GraphRAG [108] provides an overview of the key methodologies and highlights how leveraging structural relationships among entities enables more precise and comprehensive retrieval. By capturing relational knowledge within the graph, it facilitates more accurate and context-aware responses. Recent research shows that balancing intrinsic LLM reasoning with external graph information,

as demonstrated by GraphRAG-FI [137], enhances knowledge integration while lowering noise and hallucinations.

GraphRAG [137] is preferred over standard RAG when dealing with large document collections where the queries require a global thematic understanding. HybridRAG [134] is preferred in an enterprise setting where the queries are posed over both structured relational and unstructured text data. Standard RAG [4] is still preferable if the graph construction cost is limited or the knowledge is mainly unstructured. GraphRAG has substantial limitations on scalability, despite its benefits for global query understanding. Peng et al. [108] point out that the performance is directly dependent on the quality of the automatically constructed graph index; the static topology of the graph index cannot adapt dynamically to various query structures, and GNN-based retrievers may suffer from modality mismatch between graph and query embedding spaces, thus restricting GraphRAG's use in large-scale dynamic environments.

8.1.3. KG for interpretability

KGs exhibit the knowledge structurally and serve as a tool to interpret and validate the LLM outputs. Interpretability often refers to validating or explaining the decision-making process or the reasoning path of an LLM. KGs excel in this, as they provide good interpretability and a reasoning path for the output generated. LLMs are generally black box models, making it hard to understand their decision-making process. When a model prediction is done, it's mapped to the graph entities and relations, allowing fact checking and explanation, thereby providing transparency for every answer generated.

8.2. LLM – augmented KGs

This paradigm enhances the KGs using LLMs rather than KGs improving LLMs. Fig. 16 shows how this paradigm shift helps in enhancing KGs, as knowledge graphs often suffer from incomplete information; it enhances KG construction, completion, and generation. This can be done in multiple ways:

8.2.1. KG embedding with LLM augmentation

LLMs such as encoder-only LLMs are very good at encoding the given text to generate textual representations for entities and relations. When textual representations are available for graph nodes, these embeddings help in tasks like entity alignment, relation classification, and link prediction.

8.2.2. KG completion and construction

Knowledge graphs are incomplete by nature, and this is the major drawback it faces. LLMs are pre-trained on massive text corpora and, using their existing textual context, they help predict missing entities or relations. They generate new facts by doing KG completion and extract entities and relationships from texts to construct a new graph and facilitate tasks like KG-to-text-generation. The KG construction involves LLMs to perform tasks like entity extraction using named entity recognition, relation extraction, and coreference resolution. These techniques are used to build a KG by extracting relevant entities and relationships from the text document. Recent research presents an organized engineering method for creating enterprise knowledge graphs that makes use of LLMs to improve ontology design and minimize manual knowledge engineering effort [138].

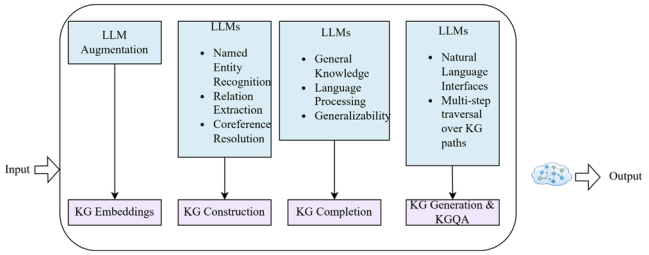


Fig. 16. LLM-augmented KG paradigm.

8.2.3. KG generation and question – answering

LLMS can convert a graph into natural language, enabling KG-to-text-generation for summarization and vice versa for knowledge acquisition. KGQA extracts answers for user queries based on the structured facts in knowledge graphs. LLMs serve as entity/relation extractors by extracting the entities and relationships in the user query and retrieving facts from a KG. LLMs are also good at managing databases and retrieving information from graph databases. For example, the GraphDB framework detects the entities when the user asks a query, and the model traverses through the KG up to N hops, where relevant subgraph paths are extracted. These paths thereby become the evidence context for LLM, and then the LLM summarizes an answer based on those reasoning paths. In addition, BERT, which has become a milestone in natural language processing, and GPT models excel at the generation and optimization of queries, thereby enhancing query performance.

Standard benchmarks for evaluating KG-LLM question answering systems are WebQuestionsSP [151] consisting of 4,737 natural language questions answerable over Freebase with one and two hop reasoning, and KQA Pro [150] offering about 120,000 diverse complex questions with explicit compositional programs over Wikidata to support evaluation of multi-hop inference, attribute comparison and set operations. These benchmarks allow for normalized comparison of KG-LLM integration strategies on structured reasoning tasks.

8.3. Synergized LLMs+KGs

This approach enables an integrated framework where LLMs and KGs are both integrated in a loop, enhancing each other's capabilities. Fig. 17 shows a joint mechanism is used where LLMs use structured knowledge from KGs to enhance their reasoning and reduce hallucinated answers by generating factual answers retrieved from KG, and on the other hand, KGs excel at response generation and summarization using contexts of LLMs.

8.3.1. Unified knowledge representation

To create hybrid knowledge representations, neural text representations and symbolic graph structures are unified in synergized models. In this way, the entities can have both graph embeddings and contextual language embeddings. This maintains the logical structure while enabling better handling of hidden entities.

8.3.2. Neuro symbolic reasoning

This provides a better way to create AI systems that combine neural

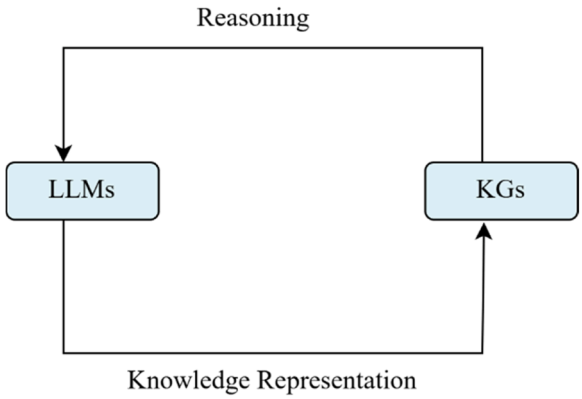


Fig. 17. Synergized LLMs+KGs paradigm.

learning with LLMs and symbolic, structured, multi-hop reasoning with KGs. Wherein LLMs excel at language understanding and KGs work with hard logical constraints and multi-hop relational paths. The theoretical foundation of the KG-LLM integration paradigms is neuro-symbolic reasoning, considered in this section. The core problem it tackles is that pure neural models, including LLMs such as BERT [10] and GPT [1], are good at pattern recognition and language understanding but hallucinate at verifiable multi-hop logical inference, whereas pure symbolic systems such as KGs are good at structured relational reasoning but struggle at handling natural language or generalizing beyond explicitly stored facts. Rocktäschel and Riedel [152] proposed Neural Theorem Provers (NTPs) by replacing symbolic unification in Prolog-style backward chaining with differentiable neural computations over KG entity and relation embeddings, establishing the basic principle of this field. This work proves that logical rules can be learned end-to-end from incomplete knowledge bases by combining symbolic KG structure with neural representation learning.

DeLong et al. [111] identified three dominant neuro-symbolic architectures for KG reasoning: neural theorem provers [152], graph neural networks with logical constraints [138], and hybrid inference systems that alternate between neural generation and symbolic KG validation. Each of these maps directly to an integration paradigm in this survey: neural theorem proving for KG-enhanced LLM pre-training [133, 136], GNN-based approaches for dynamic KG retrieval [135], and hybrid inference for the synergized LLMs+KGs paradigm [131]. This connection shows that the KG-LLM integration domain is a large-scale implementation of neuro-symbolic principles enabled by the representational power of modern LLMs.

This unification helps to reduce hallucinations in LLMs and provide more robust multi-hop reasoning, develop multi-modal LLMs for KGs, and enhance bidirectional reasoning with integrated LLMs and KGs [138].

8.4. Comparative analysis of LLM-KG integration strategies

The integration strategies compared in Table 12 vary along three basic dimensions: knowledge update flexibility, depth of reasoning, and cost of computation.

Static KG-enhanced methods like ERNIE [133] and KEPLER [136] incorporate factual knowledge into the LLM parameters during pre-training. This leads to a deep integration but makes the approach completely inflexible, demanding a full retraining cycle for any knowledge updates. Dynamic approaches such as GNN-RAG [135] do so by retrieving KG subgraphs at inference time, but the quality of retrieval is still limited by KG completeness. LLM-augmented KG methods [112, 132] reverse the enhancement direction by completing sparse KGs using LLMs, but risk hallucinated triples that deteriorate downstream quality. Synergized approaches [131,111] achieve the deepest reasoning depth through bidirectional enhancement, but require the most complex pipeline and the highest computational resources. Practical alternatives are offered by GraphRAG [137] and HybridRAG [134]. GraphRAG is designed for global thematic queries through community-based summarization, and HybridRAG is designed for an enterprise mixed-query environment through vector-graph fusion. Importantly, neither falls neatly into the three paradigms of Pan et al. [131]; they represent practically motivated retrieval implementations that extend the KG-enhanced LLM paradigm. All strategies share a common trade-off: the deeper the knowledge integration, the higher the accuracy, but the lower the updating flexibility and the higher the cost.

9. Applications of KGs

Knowledge graphs, after being introduced by Google, are being widely used in miscellaneous domains today, and full-fledged research is being conducted using KGs. Studies on applying KGs into diversified domains and in practical systems are at present the research intention

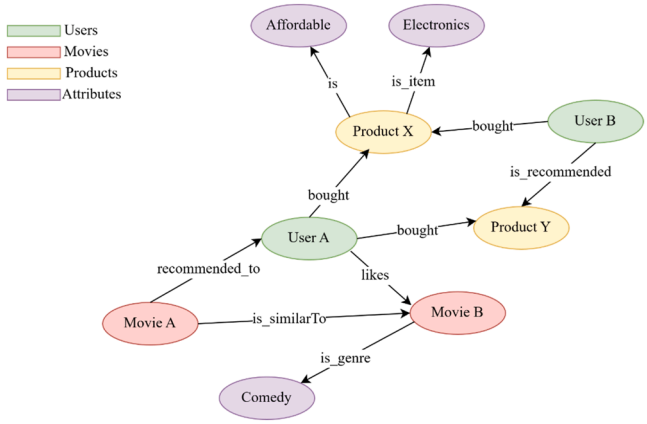


Fig. 18. KGs in Recommender Systems.

for many researchers. KGs are mostly applied in recommender systems, question answering tasks, and information retrieval systems. Knowledge graphs can be used for a specific domain like education, medical, financial, legal and cybersecurity. Below are the summarized applications of knowledge graphs.

9.1. KGs in recommender systems

There is a tremendous volume of online content available, and using KGs in recommender systems can solve the problem of data sparsity. Recommender systems shown in Fig. 18 are usually an information filtering system that predicts the preferences of the customer based on previous ratings. Embedding-based approaches leverage knowledge graph embedding (KGE) algorithms to preprocess the knowledge graph and learn vector representations of entities. These learned embeddings can then be incorporated into recommendation frameworks, allowing the systems to utilize the structural information present in the knowledge graph and thereby improve recommendation performance [145]. Multi-modal knowledge graphs can also be used, which include images that can be modeled using entity information aggregation and entity relation reasoning. To check how the information is propagated and translated, a graph attention mechanism was also employed, and many practical experiments are also underway to improve the quality of recommender systems.

9.2. KGs in information retrieval systems

IR systems like search engines are one of the best examples to showcase how knowledge graphs are useful for retrieving the information correctly, as shown in Fig. 19. Traditional systems often relied on keyword matching, which often led to incomplete or irrelevant results. In contrast, KGs understand the context and relationships between the entities and retrieve ground truth answers, thereby reducing hallucinations by their semantic understanding [146]. The function of knowledge graphs in information retrieval systems is illustrated in Fig. 19, which shows how precise answer retrieval is made possible by structured semantic relationships between queries, documents, named entities, and concepts.

9.3. KGs in medical domain

As the electronic health care data is growing tremendously, the need for factual and consistent information retrieval is much needed. KGs help in the medical field by identifying rare diseases and detecting complications, wherein a clinical knowledge graph unifies the clinical and molecular data to support decision-making [147]. UMLS is a data source that includes a semantic network where health care systems can

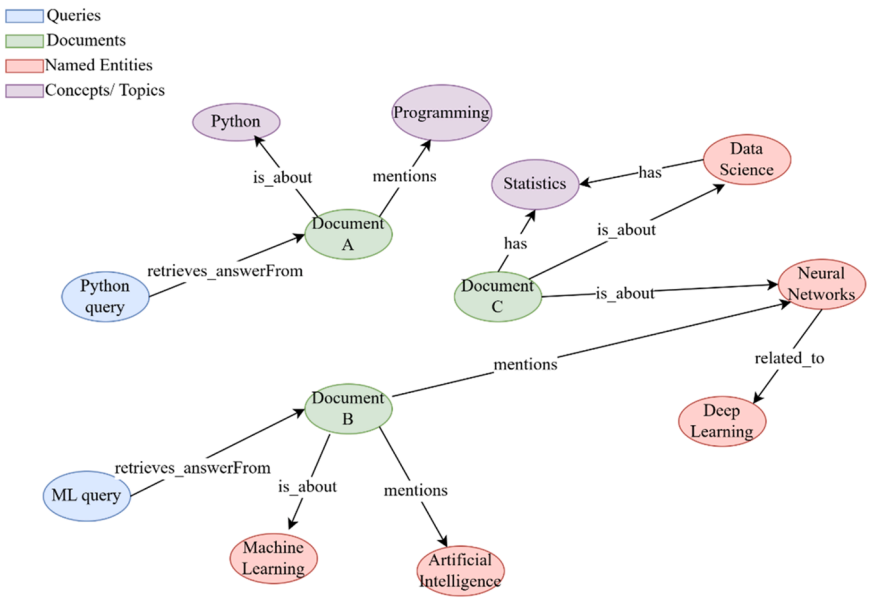


Fig. 19. KGs in Information Retrieval Systems.

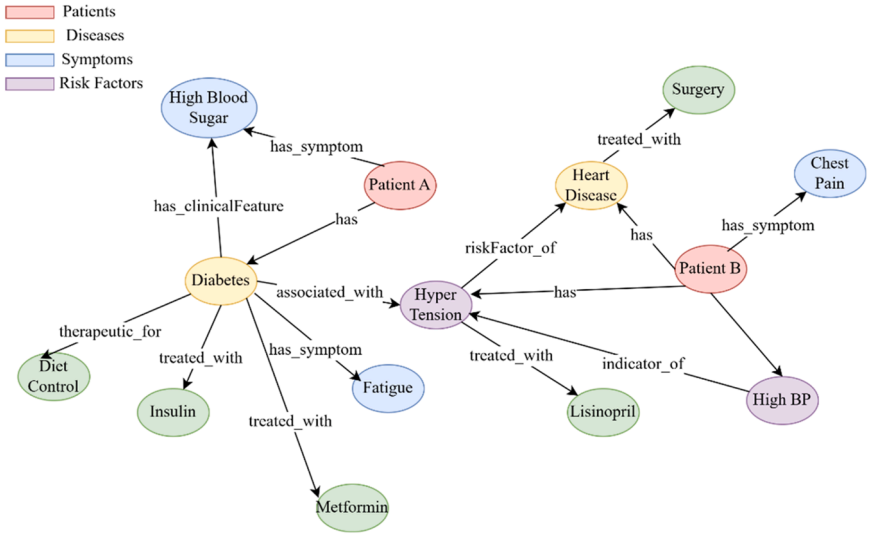


Fig. 20. KGs in Medical Domain.

connect complex biomedical data that can be reasoned in a structured manner. It is necessary to retrieve factual answers in this era of explosively growing online data, thereby reducing hallucinations. KGs assist in drug discovery and therapy selection, thereby giving new insights into the field of medicine. Graph-based transformer model (GraphBERT) captures multimodal signals like text and imaging. It works best for reasoning and inference in huge medical knowledge graphs. Fig. 20 shows the medical KG connecting patients, diseases, symptoms, and medications in a structured representation that helps in clinical decision making and patient monitoring.

9.4. KGs in fraud detection (Finance)

KGs, due to their complex patterns and relationships, enable any fraudulent activity by identifying unusual patterns like a sudden hike in the transactions with no historical connection. KGs discover hidden patterns between entities and relationships by learning relational embeddings. Graph anomaly detection is how it finds unusual patterns in data and abnormalities. If a particular entity A is related to a fraud ring B, this can be identified using link prediction. Community detection algorithms are also applied to KGs for domains such as banking, insurance, and many others to increase detection accuracy and reasoning. The graph shown in Fig. 21 connects users to their accounts, which initiate transactions. A transaction can be flagged as fraud if a merchant entity is related to fraud rings using link prediction.

9.5. KGs in customer relationship management (CRM)

To have a good relationship between the customer and the company, CRM is essential to manage interactions. KGs can integrate the complete data of a particular customer, like their purchase history, social media activity, interests, and preferences, into an overall view, which helps the companies to better understand their customers, thereby maintaining a healthy relationship with them. KGs help companies investigate the up-

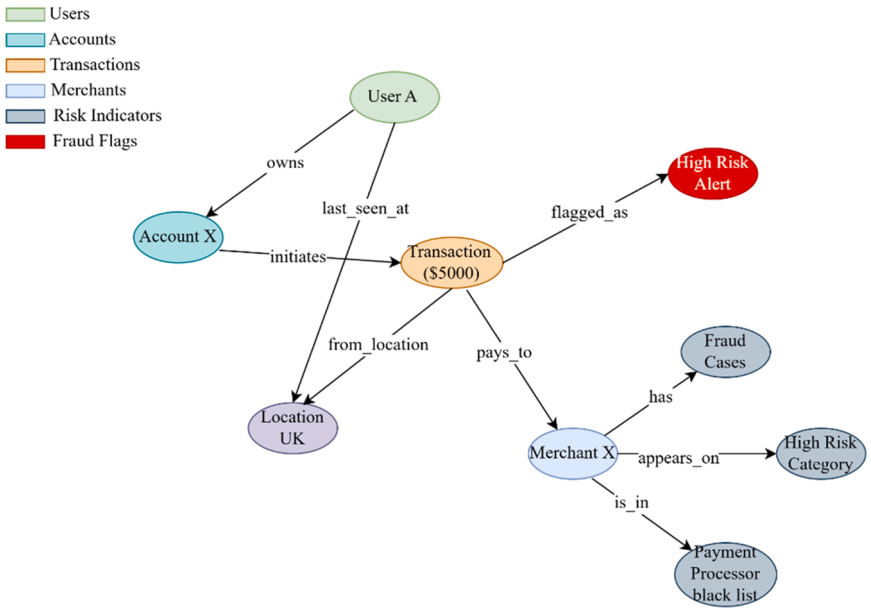


Fig. 21. KGs in Fraud Detection (Finance).

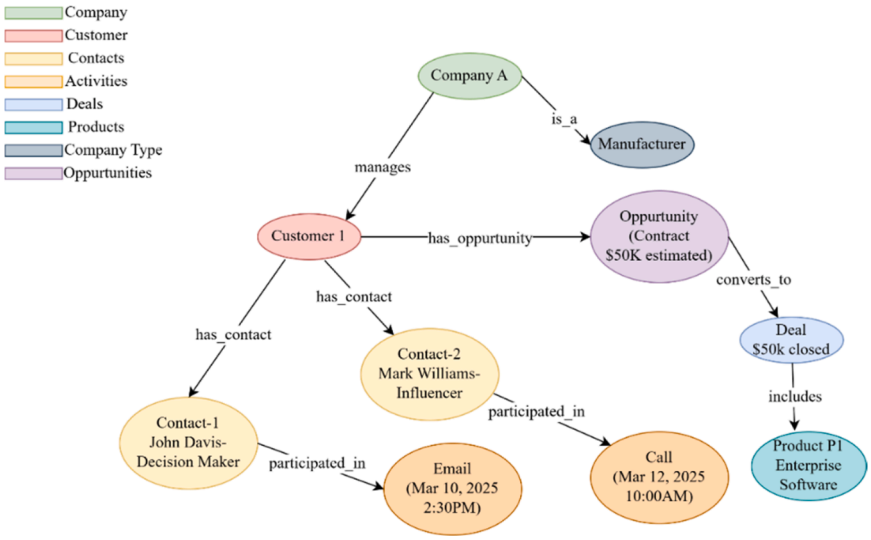


Fig. 22. KGs in Customer Relationship Management.

sell and cross-sell opportunities (like Customer A, who bought an item X, had also bought an item Y) by having strong relationships between products and customers [148]. In another scenario shown in Fig. 22, a company manages customers (stakeholders), and each customer is associated with contacts, sales opportunities that convert to deals. Project associations enable deals that include products. By linking all these entities together, CRM KGs support customer engagement strategies, enhancing sales forecasting.

These applications are just to name a few, but KGs excel at many more industrial use cases like Supply chain management, Fake news detection, biomedicine, legal, and educational systems, thereby improving the faithfulness and reducing the hallucinations.

10. Evaluation metrics used for KGs

The evaluation of KG-LLM systems is fundamentally different from the traditional evaluation of KGs, which is primarily based on link prediction metrics. The KG-LLM systems are developed to support different tasks, including KG construction, KG completion, natural language generation and retrieval, each of which is associated with its own evaluation criteria. Table 13 organizes the metrics used in this survey according to applicable scenario and task, and provides practitioners a systematic reference for selecting appropriate metrics for their specific KG-LLM evaluation needs.

10.1. Knowledge graph construction metrics

1. Accuracy: It measures the percentage of correctly predicted outcomes out of all the total outcomes. As defined in (1), it evaluates how accurate the response generation of a model is, especially in entity recognition, question answering tasks, and summarization tasks. Accuracy is critical in entity extraction in KG-LLM pipelines, where wrongly classified entities propagate errors to downstream KG triples and degrade LLM grounding at inference.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Table 13
Classification of evaluation metrics according to applicable scenarios and tasks in KG-LLM systems.

Scenario	Tasks	Metrics	KG-LLM Specific Application	References
KG Construction	NER, relation extraction, triple quality evaluation	Accuracy, Precision, Recall, F1 Score, MSE	Precision, Recall, and F1 in entity extraction directly determine the completeness and accuracy of KG nodes and relations — low scores indicate the noisy or incomplete KGs that degrade LLM factual grounding during inference	[139,140]
KG Completion and Reasoning	link prediction, entity alignment, multi-hop reasoning	Hits@k, Mean Reciprocal Rank (MRR)	Hits@k and MRR are metrics for evaluating if the KG completion model correctly ranks the missing entities and relations that the LLM retrieves during GraphRAG or HybridRAG inference—directly measuring the quality of the structured knowledge available to the LLM for multi-hop reasoning.	[139,143]
Natural Language Generation	KG-grounded LLM text output quality	BLEU, ROUGE, Faithfulness, Hallucination Rate, Exact Match	Faithfulness and Hallucination Rate evaluate whether LLM outputs are grounded in retrieved KG triples, isolating KG-grounding failures from general language model errors. BLEU and ROUGE evaluate the lexical quality of KG-informed responses, but do not capture factual consistency with source KG triples.	[141,142]
Retrieval and Ranking	GraphRAG and HybridRAG retrieval quality	nDCG, Answer Relevancy	nDCG and Answer Relevancy evaluate the relevance ranking of KG subgraphs and community summaries retrieved by GraphRAG and HybridRAG to the user query, directly measuring the effectiveness of graph-based retrieval for supplying accurate context for LLMs.	[144,108, 134]
System-level	Overall KG-LLM pipeline performance	Latency, Helpfulness	Latency is the total overhead of KG traversal, subgraph retrieval and LLM generation – finding bottlenecks specific to graph-augmented inference; Helpfulness measures user-perceived quality of KG-grounded responses beyond automated metrics.	[108,139]

Where a true positive (TP) is a correctly predicted outcome, a true negative (TN) is the correctly predicted negative outcome, a false positive (FP) denotes a negative outcome which is predicted to be positive, and a false negative (FN) is a positive outcome predicted to be negative.

2. Precision: It is the percentage of correctly classified outcomes out of all the positive outcomes. Basically, it tests how many of the instances predicted as positive were actually positive [139]. As defined in (2), Precision is used to measure the relevant outcomes correctly in tasks like entity alignment and entity matching. In KG construction, high precision in Named Entity Recognition ensures that only the correctly identified entities are added as nodes, preventing LLMs from retrieving hallucinated facts from a noisy KG.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \tag{2}$$

3. Recall: This is also referred to as Sensitivity. It is the percentage of all the true positive outcomes from the total actual positive outcomes in the dataset. As defined in (3), Recall tests the model's ability to identify how many true positive outcomes the model identified correctly [140] in tasks like entity alignment and matching. In KG-LLM systems, low recall during relation extraction leads to incomplete graphs with missing edges, directly limiting the multi-hop reasoning paths available to the LLM during GraphRAG or HybridRAG retrieval.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{3}$$

4. F1- Score: As defined in (4), it is the integration of precision and recall calculated by the harmonic mean of precision and recall. It is useful when dealing with imbalanced datasets to undergo a balanced evaluation of the model's performance. This is also referred as F-Measure [139]. F1 is the main metric for assessing NER and relation extraction tools (Table 7), and directly influences the structural completeness of the KG queried by LLMs at inference time.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

5. MSE (Mean Squared Error): MSE quantifies the model accuracy by measuring the average of squares of residuals, which is the difference between actual and predicted values. It is used in regression tasks. MSE is used to evaluate embedding-based KG completion models such as TransE and ComplEx in KG-LLM systems. A lower MSE value for entity/relation embeddings implies higher accuracy in link prediction and better KG retrieval for LLMs.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 \tag{5}$$

Where Y_i is the actual value, $\hat{Y}_i$ is the predicted value of the ith observation.

10.2. Knowledge graph completion and reasoning metrics

1. Hits@k: It is the count of how many times the correct answer is ranked in the top-k positions against a pool of negatives. It is widely used in link prediction and response generation tasks to evaluate the performance of the model. In a GraphRAG or HybridRAG pipeline, Hits@k directly measures if the KG completion model correctly ranks the missing entity that the LLM needs to retrieve among the top-k candidates – low Hits@k means the LLM receives incorrect or incomplete subgraphs as context.

$$Hits@k = \frac{H_k}{N_{query}} \tag{6}$$

Where H_k is the count of times the right answer is in the top-k and N_{query} is the total number of queries.

2. MRR (Mean Reciprocal Rank): MRR is used in information retrieval tasks like question answering and recommendation systems. MRR assesses the rank of the first correct document retrieved for a set of queries [143]. This is commonly used for tasks like link prediction and KG completion. In KG-enhanced LLM systems, MRR reflects how fast the correct entity/relation is surfaced when traversing the graph, and thus directly affects the factual correctness of the LLM's generated output in multi-hop QA tasks.

$$MRR = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{rank_i} \tag{7}$$

Where N is the number of queries and $rank_i$ is the rank for the first correct result for query i.

10.3. Natural language generation metrics

1. ROUGE (Recall-Oriented Understudy for Gisting Evaluation): ROUGE is based on precision, recall, and F1-score for tasks like text-summarization. It evaluates the quality of model-generated text by comparing the reference summaries to assess how apt the summary generated by the model relates to the reference summaries [141]. For KG-LLM, ROUGE measures how well the KG-grounded summary generated by the LLM covers the important entities and relations in the retrieved subgraph, but does not penalize factually incorrect outputs that are lexically close to the reference. The key ROUGE metrics are: a) ROUGE-N: It calculates the n-gram overlap between the generated summary and the reference summary.

$$ROUGE - N = \frac{Number\ of\ matching\ n - grams}{Total\ n - grams\ in\ the\ reference} \tag{8}$$

b) ROUGE-L: It measures the similarity by taking the longest common subsequence in the same order between the model-generated summary and the human-given reference summary.

$$ROUGE - L = F_\beta = \frac{\left(1+\beta^2\right).P.R}{\beta^2.P + R} \tag{9}$$

Where P is precision, R is recall, and β is typically set to 1.

2. BLEU (Bilingual Evaluation Understudy): BLEU uses precision to calculate the number of words that overlap between machine-generated text and human-referenced text. It calculates the count of combinations of n words, basically precision [142]. In KG-to-text generation tasks, BLEU measures the accuracy of an LLM in converting structured KG triples to natural language, but the lexical focus means that it can reward fluent hallucinated outputs that contradict the source KG triples.

$$BLEU = BP.\exp\left(\sum_{n=1}^{N} w_n log\ p_n\right) \tag{10}$$

Where,

BP is the Brevity Penalty, which means a penalty is imposed if the model generates smaller candidate sentences w_n are the n-gram weights p_n is the precision for n-grams

3. Faithfulness: Faithfulness is probably the most KG-LLM-specific metric in this section – it directly measures the extent to which the generated answer from the LLM is grounded on the retrieved KG triples, and is the main metric for success of KG-grounding in systems like GraphRAG [137] and HybridRAG [134].

$$Faithfulness = \frac{N_{faithful}}{N} \tag{11}$$

Where N is the total number of responses, and $N_{faithful}$ is the number of responses that were factually correct.

4. Hallucination Rate: In KG-LLM systems, the failure of KG grounding is quantified by the hallucination rate, i.e., the rate at which the LLM generates a response that is contradicted or unsupported by the retrieved KG subgraph, which is the main failure mode these systems are designed to avoid.

$$Hallucination\ Rate = \frac{N_{hallucinated}}{N} \tag{12}$$

Where N is the total number of responses, and $N_{hallucinated}$ is the number of responses that were hallucinated.

5. Exact Match (EM): The Exact Match score is the proportion of predicted perfect matches between the model-predicted texts and the reference texts. The value ranges from 0 to 1. The higher the value, the stronger the predicted text that matches the ground truth text. EM measures whether the LLM's final answer is exactly the same as the ground truth entity stored in the KG, so it is a strict but direct measure of the correctness of KG retrieval. In KG-based QA benchmarks such as WebQuestionsSP [151] and KQA Pro [150].

$$Exact\ Match = \frac{PEM}{N_{pred}} \tag{13}$$

Where PEM is the proportion of the perfect matches, and N_{pred} is the total number of predictions.

10.4. Retrieval and ranking metrics

1. nDCG (Normalized Discounted Cumulative Gain): nDCG measures the ranking quality of how well the model structures the results by their relevance to the reference texts. In the ranked list, a higher rank is given to the most relevant document. To normalize the score, it ranges from 0 to 1 [144]. In GraphRAG pipelines, nDCG is used to evaluate if the top KG community summaries or subgraphs are ranked at the top for a given query, as a direct evaluation of the graph-based retrieval quality before the LLM generation.

$$nDCG@k = \frac{DCG@k}{IDCG@k} \tag{14}$$

2. Answer Relevancy: It calculates the semantic similarity between the generated answer and the original query. It is particularly appropriate for the KG-LLM evaluation because unlike BLEU or ROUGE, it does not need a reference text – it evaluates if the LLM's KG-grounded response addresses what the user asked semantically, regardless of the exact wording [108,134].

$$\text{Answer Relevancy} = \frac{1}{N}\sum_{i=1}^{N} \cos\left(E_{gi},\ E_o\right) \tag{15}$$

Where N is the number of generated questions, E_{gi} is the embedding of the ith generated question, E_o is the embedding of the original question/query

10.5. System-level metrics

1. Total Latency: It captures the end-to-end time taken, from the time when the user submits a query to the response generated by the model as a final output. It measures the total response delay in all the processing steps.

$$Total\ Latency = t_{end} - t_{start} \tag{16}$$

Where t_{start} is the time when the user has submitted the query, and t_{end} is the time when the response was generated by a model.

Latency is calculated at all processing steps and can be written as,

$$Total\ Latency = t_{retrieval} + t_{reasoning} + t_{generation} + t_{overhead} \tag{17}$$

Latency In KG-augmented systems, latency is higher than standard LLM inference, due to the extra overhead of graph traversal, subgraph retrieval and embedding lookups. It is an important metric for production deployment of GraphRAG [137] or HybridRAG [134] systems.

2. Helpfulness: It gauges whether the answer generated by the model is useful for the user. It checks whether the response generated was complete, relevant, and informative i.e., whether the structured KG grounding actually made the response more useful to the user than a plain LLM response in KG-LLM systems. This is generally measured using a Likert-scale rubric system.

$$\text{Mean helpfulness score} = \frac{1}{N}\sum_{i=1}^{N} H_i \tag{18}$$

Where N is the number of total responses and H_i is the average helpfulness score for response i.

Evaluation metrics in KGs ensure that the constructed graph is semantically correct and computationally efficient for real-world applications. Faithfulness and hallucination rate are emerging as important cross-cutting metrics across all scenarios, measuring the reliability of KG grounding in LLM outputs [108,137].

Although the metrics are available, their application to KG-LLM systems presents unique challenges that are not faced in standard NLP evaluation. First, KG-LLM systems include several interdependent components, such as KG construction, retrieval, and generation, making it difficult to determine which component is causing performance degradation when end-to-end metrics like Hits@K or BLEU are used [139]. Second, Metrics such as BLEU and ROUGE measure lexical overlap with reference text but often fail to capture faithfulness to the grounding KG, where fluency is seen as a primary concern rather than hallucination. Third, metrics such as MRR and Hits@k, often used for link prediction, evaluate KG completion in isolation but fail to measure whether the new triples improve LLM performance, which can create a disconnect between KG-level and system-level evaluation [132]. These challenges highlight the need for unified assessment tools to measure the KG quality and LLM performance when unified.

11. Conclusion and future research directions

11.1. Conclusion

Large language models (LLMs) have recently evolved in a variety of domains, particularly for language-related tasks like summarization and question-and-answer sessions. In this work, we aim to present a survey on unifying LLMs and KGs. We started by discussing the basic architectures of LLMs, including encoder-only, decoder-only, encoder-decoder, and multimodal models. A comprehensive taxonomy of Knowledge Graphs was discussed with a focus on multimodal KG construction and healthcare-specific KG construction incorporating clinical NLP pipelines, medical ontologies, and EHR-based graph building. The KG construction paradigms were investigated end-to-end through data acquisition, ontology design, entity and relation extraction, knowledge fusion, and graph reasoning. After a thorough analysis of the three integration strategies — KG-enhanced LLMs, LLM-augmented KGs, and Synergized LLMs+KGs — GraphRAG and HybridRAG were highlighted as new retrieval-augmented frameworks that improve multi-hop reasoning and factual grounding. The practical significance of this integration was further confirmed by evaluation metrics and applications in the fields of healthcare, finance, recommender systems, and information retrieval.

11.2. Future research directions

However, there are still a few unresolved issues, such as knowledge injection into black-box LLMs, cross-modal alignment, reasoning over incomplete graphs, and dynamic knowledge updating without retraining. For this field to advance, it will be essential to address these issues through lightweight KG construction, ongoing learning, and standardized benchmarks. The convergence of the generative power of LLMs and the symbolic accuracy of KGs, in our opinion, represents a critical step toward developing AI systems that are not only strong but also factually based, comprehensible, and reliable. Based on the analysis in this survey, we identify five priority directions for future research at the KG-LLM intersection:

11.2.1. Dynamic knowledge graph updates

One major open challenge is how to continuously update KGs without requiring full retraining of integrated LLMs. Static KG-enhanced approaches such as ERNIE [133] and KEPLER [136] embed KG facts during pre-training, making them computationally prohibitive to update. Future work should explore incremental update mechanisms that only propagate changed triples through the adapter layers or retrieval cache updates. GDELT [66] and ICEWS [67] are temporal KG frameworks that could serve as baseline architectures for integration with dynamic LLMs. Future benchmarks should include time-stamped KGQA datasets that explicitly test performance on facts that change over time.

11.2.2. Cross-modal alignment in MMKGs

Vision-language models have seen progress, but the modality gap between visual and textual embedding spaces persists [149], leading to systematic misalignment in multimodal KG-LLM systems. Future work should develop cross-modal alignment objectives that explicitly minimize geometric distance between paired modality embeddings, combined with uncertainty-aware entity linking that flags low confidence correspondences for human review. WIKIDiverse [115], VisualSem [88] and UKnow [89] provide some initial evaluation points, but there is a need for more diverse domain-specific benchmarks for cross-modal MMKG evaluation.

11.2.3. Reasoning over incomplete graphs

KG completion methods [112] that work well on dense knowledge bases are at risk of hallucinating relationships for sparsely documented entities — particularly in rare disease healthcare settings [52]. Future work should develop uncertainty-aware KG completion, in which confidence scores are assigned to predicted triples and low-confidence completions are surfaced to domain experts before integration. KQA Pro [150] and WebQuestionsSP [151] benchmark KG-based QA, but neither particularly investigates the performance degradation caused by KG incompleteness, which is a gap to be filled in future benchmark design.

11.2.4. Standardized evaluation benchmarks

There is no unified benchmark to evaluate the whole KG-LLM pipeline from construction to integration to application. Existing benchmarks evaluate individual components, such as KG construction [139], link prediction [143], and text generation [141,142], but do not evaluate the end-to-end pipeline jointly. Future work needs to develop evaluation frameworks that jointly evaluate the quality of KG construction, retrieval faithfulness, generation accuracy, and hallucination rate for fair comparison of GraphRAG [137], HybridRAG [134], and synergized approaches [131].

11.2.5. Ethical issues and bias propagation

Bias transmission between KGs and LLMs is an underexplored but critical ethical concern. Encyclopedic KGs like Wikidata [47] and DBpedia [46] are known to inherit societal biases by over-representing entities from certain cultures and geographic regions. When such biased KGs are used to ground LLM reasoning through KG-enhanced integration [131,133], these structural biases are further amplified through the LLM's generative capabilities. On the other side, LLMs used for KG construction [112] could generate culturally biased triples that get permanently embedded in the graph. Future research should develop mandatory bias auditing pipelines for KG-LLM systems pre-deployment. This is particularly important in healthcare settings [129,130], where biased clinical KGs could result in unfair diagnostic recommendations.

CRediT authorship contribution statement

Qurratul Aine Syeda: Writing – review & editing, Writing – original draft. **Ravinder Reddy B.:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

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